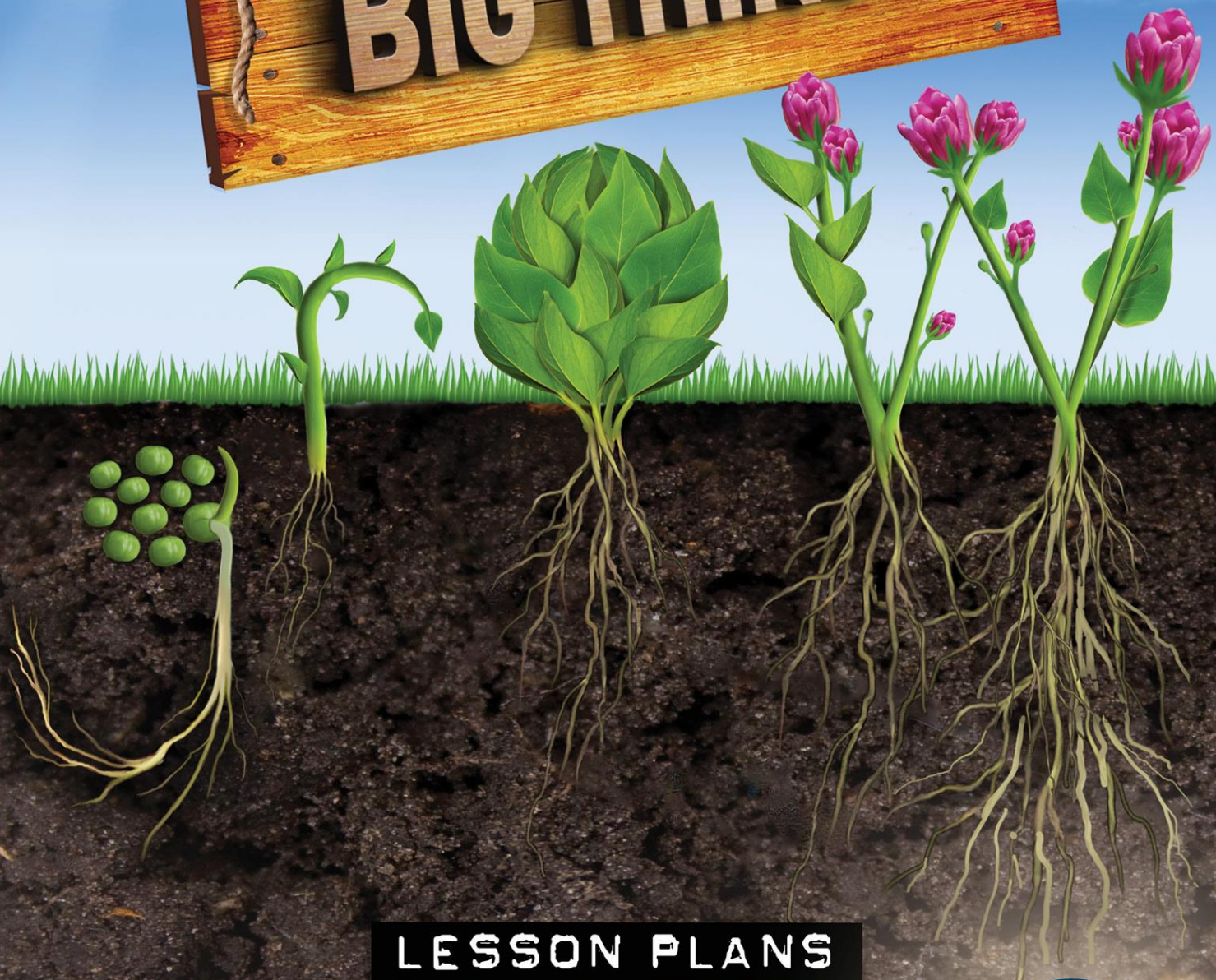


**FROM LITTLE THINGS
BIG THINGS**



LESSON PLANS

The core activities in this unit total about 24 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Part 1 What is life?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	Living or non-living?	ACSSU150	Engage & Evaluate	Students explore a variety of stimulus material to develop an understanding of the terms living and non-living.	Optional
			Digital		
			Short		
1.2	Characteristics of life	ACSSU150	Explore, Explain & Elaborate	Students observe sultanas in carbonated water to simulate a possible new life form and use a second simulation to explore movement as a possible characteristic of life. Students are then introduced to the more formalised characteristics of life and investigate ambiguous situations of life to consolidate their understanding of the term living.	Core
		ACSHE135	Hands-on & Digital		
		ACSHE136	Long		
1.3	Plant life	ACSSU150	Explore & Explain	Students grow broad bean seeds and record observations of their growth throughout the unit. Experimental design and data collecting skills are revised, leading later to Activity 5.8 where students will be reporting conclusions.	Core
		AC SIS140	Hands-on		
		AC SIS141			
1.4	Is Sammy alive?	AC SIS144	Long	This activity is based on an animation of the life of Sammy, whose body parts have progressively been replaced. Students participate in group discussions to decide at what stage Sammy is no longer alive.	Core
		AC SIS145			
		AC SIS148			
1.4	Is Sammy alive?	ACSSU150	Elaborate	This activity is based on an animation of the life of Sammy, whose body parts have progressively been replaced. Students participate in group discussions to decide at what stage Sammy is no longer alive.	Core
		ACSHE135	Digital		
		ACSHE136	Short		

Part 2 What are cells?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	Cell theory	ACSSU149 ACSHE134 ACSHE226 ACSHE136	Explore & Explain	Students use written material and digital interactives to study the historical and technical development of microscopes and learn about the capabilities of each microscope. In the Find out more students see detailed images of insects, diatoms, blood cells and wood taken with a scanning electron microscope.	Core
			Classroom & Digital		
			Medium		
2.2	How small can things be?	ACSIS141 ACSIS144	Explore & Explain	Students learn to use and convert units as they explore the concepts of size, measurement and scale diagrams; focusing on units used for microscopic objects. An animation helps visualise the relative size of objects and the appropriate units to use in measurement	Core
			Classroom & Digital		
			Long		
2.3	Making little things bigger	ACSSU149 ACSIS141	Explore, Explain & Evaluate	Students learn the parts and functions of the microscope and how to set up and focus prepared slides.	Core
			Hands-on & Digital		
			Medium		
2.4	Observing cells	ACSSU149 ACSHE134 ACSIS141	Explore & Explain	Students use photomicrographs of cells to familiarise themselves with basic cell parts, before exploring a virtual microscope to investigate how magnification, focusing and light aperture affect the image.	Core
			Classroom & Digital		
			Short		
2.5	Microscope skills	ACSIS141	Explore & Explain	Students further explore the use of the microscope and learn about the inversion of images and resolution. They learn how to make measurements using minigrids.	Optional
			Hands-on		
			Medium		
2.6	Looking at living cells	ACSSU149 ACSSU150 ACSIS141	Explore & Explain	Students follow instructions to prepare a wet mount of onion skin cells. They focus under low and high magnification, before staining the cells and comparing the effect on the image seen.	Core
			Hands-on & Digital		
			Medium		

Part 2 What are cells? (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.7	Drawing cells	ACSIS144	Explain	The class will view a variety of biological diagrams drawn by scientists and students. Students will learn about drawing biological diagrams and draw labelled cells from photomicrographs.	Core
			Classroom		
			Medium		

Part 3 What goes on inside a cell?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	What's inside a cell?	ACSSU149 AC SIS144	Explore, Explain & Elaborate	Students view stylised cell diagrams to explore the classification of cells and use their findings to classify cells.	Core
			Digital		
			Short		
3.2	What makes cells function?	ACSSU149	Explore, Explain & Elaborate	A jigsaw activity in which groups explore the structure and function of different cell organelles and regroup to share findings. The Notebook activity introduces single-celled organisms and invites students to apply their knowledge to new situations.	Core
			Digital		
			Medium		
3.3	Yeast metabolism	ACSSU149 AC SIS139 AC SIS140 AC SIS141 AC SIS144 AC SIS145 AC SIS146 AC SIS234 AC SIS148	Explore & Explain	In this activity students will explore the metabolism of yeast by planning and carrying out a controlled experiment to investigate the effect of temperature on cell activity, then graphing the results. They will also use a microscope to observe yeast budding.	Core
			Hands-on		
			Long		
3.4	Cell division	ACSSU149 ACSSU150	Explore & Explain	Students use digital media to learn about binary fission and mitosis as methods of cell division involved in growth and reproduction.	Core
			Digital		
			Short		
3.5	Why are cells so small?	ACSSU149 AC SIS140 AC SIS141	Explore & Explain	Students carry out an investigation using agar blocks to represent different sized cells in order to study the efficiency of diffusion in different sized cells.	Core
			Hands-on		
			Medium		
3.6	Practical skills in biology		Evaluate	A practical skills test for Units 1 - 3 (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional
			Hands-on & Classroom		
			Medium		

Part 4 How can little cells get together to make big things?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	Life in water	ACSSU149 ACSSU150	Engage & Explore	Students observe organisms in pond water using microscopes.	Optional
			Hands-on		
			Medium		
4.2	Cells specialise	ACSSU150	Explore, Explain & Elaborate	Students investigate tissues using prepared slides, wet mounts, and virtual microscope slides.	Core
			Hands-on & Digital		
			Medium		
4.3	Putting it all together	ACSSU150	Explore, Explain & Elaborate	The concept of cell-tissue-organ-system-organism is explored using simulations of human and flowering plant systems.	Core
			Digital		
			Short		
4.4	Organs and systems	ACSSU150	Explore, Explain, Elaborate & Evaluate	Students investigate the structure and function of body organs and systems, focusing on the digestive, circulatory and kidney systems. The Find out more section introduces the skin as a body organ, discussing its structure and function, skin burns and skin cancer.	Core
			Digital & Classroom		
			Long		
4.5	New skin	ACSSU150 ACSHE134 ACSHE226 ACSHE136	Elaborate	Students study the life and work of 2005 Australian of the Year Professor Fiona Wood, a skin burns surgeon, using several media accounts.	Core
			Classroom & Digital		
			Medium		
4.6	Let's look inside	ACSSU150	Explore & Explain	Students dissect a squid as an example of a functioning organism.	Optional
			Hands-on		
			Medium		

Part 4 How can little cells get together to make big things? (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.7	Plant plumbing	ACSSU150	Explore & Explain	Students do experiments to study a plant vascular system and the structure and role of stomata.	Optional
			Hands-on		
			Medium		
4.8	How does a multicellular organism function?	ACSSU150	Evaluate	A formative assessment where students use packs of cards to link the components (cell-tissue-organ-system-organism) of the circulatory, digestive and excretory systems.	Optional
			Classroom		
			Short		

Part 5 Cells die, but life goes on!

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	Is death a part of life?	ACSSU149	Engage, Explore, Explain & Elaborate	Students investigate the cycle of life and death of cells by interpreting an animation of cells in intestinal villi.	Optional
			Digital		
			Medium		
5.2	A scientific breakthrough!	ACSHE134	Explain & Elaborate	Students investigate the life and work of Professor Barry Marshall by listening to an interview and exploring other media sources. They study how he made the link between the bacterium Helicobacter and stomach ulcers and cancer.	Core
		ACSHE135	Digital		
		ACSHE136	Medium		
5.3	Why the difference?	ACSSU150	Explore, Explain & Elaborate	Students compare and contrast organ systems of animals from different niches. They explore the relationships between teeth and diet, and skeletal systems and lifestyle. An interactive program is used to compare the respiratory systems of humans, frogs and fish.	Core
			Classroom & Digital		
			Medium		
5.4	Reproductive strategies in animals	ACSSU150	Explore, Explain & Elaborate	Students analyse written and visual information to make comparisons between animals with different reproductive strategies. They relate the reproductive strategies to the animal’s niche and classification.	Core
			Classroom & Digital		
			Long		
5.5	Plants reproduce too!	ACSSU150	Explore, Explain, Elaborate & Evaluate	Students study sexual reproduction in plants using digital media and hands-on experiences to learn about the structure of flowers and their function in pollination and fertilization. Students evaluate their knowledge of flower parts (formative assessment) in a click-and-drag exercise.	Core
			Hands-on & Digital		
			Long		

Part 5 Cells die, but life goes on! (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.6	But there's another way!	ACSSU150	Explore & Explain	Students investigate methods of asexual reproduction in plants. This is followed by discussion of the advantages and disadvantages of reproductive methods.	Core
			Hands-on & Classroom		
			Medium		
5.7	Presenting and interpreting experimental results	AC SIS144 AC SIS145	Explain & Elaborate	Students consolidate their skills of tabulation, graphing and drawing valid conclusions through guided activities.	Core
			Classroom		
			Medium		
5.8	So how did it grow?	ACSSU150 AC SIS140 AC SIS141 AC SIS144 AC SIS145 AC SIS146 AC SIS148	Explain, Elaborate & Evaluate	Students correlate the information collected during the growth of their bean and write a scientific report on their findings. This can be used as a summative assessment .	Core
			Classroom		
			Medium		
5.9	Sample test		Evaluate	A sample summative test for Units 1 - 5 (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional
			Classroom		
			Medium		

1.1 Living or non-living?

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

Lesson outcomes

At the end of this activity students will be able to:

- identify everyday things as living or non-living

What ideas might your students already have?

- some may not regard plants as living things
- students might attribute movement to living things
- they may not be sure how dead organisms fit in.

Key vocabulary:

Living, non-living.

Equipment list

Each CLASS will require:

- Internet access

Each STUDENT will require:

- **Notebook**

Things to consider

- Familiarise yourself with the digital resource to decide the best way to support your students' learning and manage internet access. The learning experience has been designed to put technology into the hands of the students, with the teacher taking a facilitating role.
- Students will use **Notebooking**. This could be presented in words, pictures and other media. It may be electronic or hard copy. Refer to the **Notebooking** section of the *Student Learning Professional Learning Module* for further information on getting your students started.
- What characteristics do all living things share? Is it the ability to move, reproduce, react to surroundings etc.? It is important students have adequate time to construct arguments and feel comfortable sharing their ideas as a class. Explain that you are just interested in getting a picture of what they already know.
- Do not impose formal classifications of features that make a living organism; let students use their own vocabulary. The characteristics of living things will be developed **Activity 1.2**.



Lesson plan

Step 1: Begin the lesson by showing the video Introduction to the *Characteristics of Life* (3:40) which shows a variety of living things and poses questions and contradictions to stimulate students' ideas.

Step 2: Introduce students to the first **Notebook** task 'Signs of life'. Students identify which things are living and which are non-living in the photo scenes.

Step 3: Conduct a **Think-Pair-Square-Share**.

Think – Ask students to look at their lists of living things they identified and compile a list of the characteristics that all living things have in common.

Pair – Discuss with a partner and share lists.

Square – Now meet up with another pair of students and compare lists.

Share – Now bring the class together. Invite students to share their ideas with the whole class and encourage the class to **Notebook** any new ideas that they did not think of. Listing all ideas on a whiteboard is helpful – students can act as scribes.



1.2 Characteristics of life

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

AC SIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.

Lesson outcomes

At the end of this activity students will be able to:

- recognise that the ability to move is not necessarily an indication that something is alive
- recall the seven characteristics of living things.

What ideas might your students already have?

Students should have some informal ideas from previous activities about what makes something living. These should now be expressed in more scientific terms.

Key vocabulary:

Characteristic, cell, reproduce, offspring, deoxyribonucleic acid (DNA), energy, stimuli, respond, homeostasis, trilobite, virus.

Equipment list

Each CLASS will require:

- internet access

Each GROUP will require:

- three samples of a potential new life form. Just before class prepare the following labelled samples of 'sewer lice' in 250 mL beakers, filled to 6 cm height with these mixtures:
- 'Sewer lice **larvae** from pre-treated sewage': mixture of 50% cola and soda water and 5-10 currants.



'Sewer lice **pupae** from partially treated sewage': mixture of 20% cola diluted with soda water and 5 sultanas.



- 'Sewer lice **adults** from treated sewage': soda water and 4 raisins.



- wide basin or tray
- methylated spirits
- 50 mL measuring cylinder
- 50 mL beaker
- paper dots from a hole punch machine
- forceps

Each STUDENT will require:

- safety glasses
- **Notebook**

Things to consider

- It is important to approach the first activity with excitement at a possible new discovery and encourage students to use their imaginations and direct observations, rather than preconceived ideas of what might be in the samples.
- Try out the three recipes for 'sewer lice' so that you are prepared to guide students with observations. The 'sewer lice' will continue to be active for at least half an hour.
- Bring in sewer lice samples just before use, rather than having them in the room at the start of the lesson, to keep the element of surprise. Ideally a laboratory assistant could make these up just before you require them so that the solution does not lose its fizz.



- The second activity is a clear example of inanimate objects moving by themselves and begs the question of whether movement is a characteristic of life. The dots are obviously not living, but move due to the surface tension changes between the alcohol-soaked dots and water. This is a short but exciting activity, which can be done in groups, as indicated in the *Student Guide*, or as a teacher demonstration using a larger container, such as a glass trough. A straight-sided container is better than one with curved edges.
- Students will find the video engaging but quirky. Alert students might be able to spot the two spelling errors in the video.
- Methylated spirits is flammable and should not be in the vicinity of an open flame.

Teacher content information

There is ongoing debate about what are the requirements of life. A very popular mnemonic has been MRS GREN, an acronym for:-

- Movement
- Respiration
- Sensitivity
- Growth
- Reproduction
- Excretion
- Nutrition

Students often list movement as a requirement; however it is difficult to see movement in plants unless you observe time-lapse photography or witness cytoplasmic streaming in plant cells.

In the commonly used university textbook 'Biology' by Campbell et al (2009), 'some properties of life' include:-

- Order – having a highly ordered structure
- Evolutionary adaptation – adaptations evolve over many generations by the reproductive success of those individuals with heritable traits that are best suited to their environments.
- Response to the environment – responding to environmental stimuli
- Regulation – homeostasis
- Energy processing
- Growth and development
- Reproduction

Reference –Campbell, Neil A. et al. 2009, Campbell biology, Eighth Edition (Australian version), Frenchs Forest, N.S.W. Pearson Australia

In this activity we have chosen to include the cell theory and acknowledge the following characteristics of life; more scaffolded to Year 8 understanding:-

- Cells - Living things are made up of one or more cells



- Reproduce - Living things can reproduce and produce offspring
- Energy - Living things require energy for their daily activities. Plants obtain this energy from sunlight and make sugars to store energy for later. Animals eat other organisms to get their energy.
- DNA - Living things contain an information storing molecule called DNA, and this gives them the ability to adapt and evolve (change) over generations.
- Responsive - Living things respond to stimuli in their environment.
- Growth - Living things grow and develop
- Homeostasis – Living things maintain homeostasis (e.g. water balance, stable body temperature, blood volume).

Dancing Dots

Surface tension is a property shown by liquids, such as water, which creates a 'skin' on the surface that enables some objects to float, even though they are denser than water. Surface tension is also responsible for the ability of some insects, e.g. water striders, to run on the water surface.

Surface tension is caused by the attraction (cohesion) of water molecules. A water molecule is equally attracted to other water molecules on all its sides. A water molecule on the surface of the water is attracted to other water molecules beneath it as well as next to it, but there are no water molecules above it to attract it. The water molecules at the surface are therefore pulled closer together, causing them to act like a thin elastic film.

The paper dots move on the surface of the water because the alcohol from the dot weakens the surface tension of the water near the dot. This creates an adjacent area of stronger surface tension, which pulls on the dot and causes it to move. Once the alcohol has dissipated, the surface tension regains equilibrium and movement ceases.

Movement by an organism is in response to an environmental stimulus, whereas external forces (pressure, gravity, magnetism, temperature, etc.) cause the movement of non-living things. All atoms and molecules are in constant motion. The higher the temperature the faster they move and the more frequently they collide. At absolute zero (-273°C) the movement of all particles ceases.

Reference - This activity is described in 'A Simple, Inexpensive, Dynamic, & Hands-on Exercise for Prompting Discussion of the Characteristics of Living Things' by James E. Mickle & Patricia M. Aune
<http://www.cals.ncsu.edu/plantbiology/Faculty/jmickle/pdf/mickle&aune2011.pdf>

Lesson plan

Step 1: Introduce the lesson with a brief discussion of how scientists might determine if an unknown sample contained a living organism. Outline the scenario, ensuring that students have grasped the idea of larvae developing into pupae and then adult sewer lice. Before students begin, ensure they understand the difference between an observation and an inference.



Step 2: Divide students into small groups and assign work benches. Each group should appoint a scribe to **Notebook** observations. Groups should now collect the three freshly made up samples, make observations for about five minutes and then look at the discussion questions. Conclude with an informal discussion of ideas from different groups. Stress that there are no 'right' answers here.

Suggested question/s:

- Why is this not an observation?
- Does this mean that the sewer lice are living?
- What else would you need to observe to confirm that the lice are living?

Step 3: Introduce the 'Dancing dots' as a student activity or as a teacher demonstration. Discuss as a class why the dots move, even though they are obviously not living. You may want to discuss surface tension to explain to students how movement in this non-living situation can occur.

Step 4: Ask students to design and carry out a simple fair test to find out if the movement occurs without alcohol. Circulate to check experimental designs.

Step 5: Conclude the activity by discussing whether movement alone is a good way to distinguish living things.

Step 6: Follow the link to the digital activities. The video and the rollover wheel introduce the seven characteristics of living things. Invite students to complete the **Notebook** task where they sort items into living or non-living and justify their decisions using the characteristics of life.

Suggested questions:

- What characteristics of life does this object show?
- Does a seed or fruit have to be attached to the parent organism to be alive?
- Was this once alive? How do you know?
- When is something dead?
- Was it hard to decide if some of these examples were living or not?
- Which ones were difficult?
- Can the class think of other examples that are hard to group as living or non-living?



1.3 Plant life

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- design and set up a fair test to investigate the growth of bean seeds
- understand the differences between the dependent, independent and controlled variables.

Key vocabulary:

Germination, reliability, validity, lateral roots, cotyledon, embryo, radicle, plumule, testa, scientific method, hypothesis, controlled experiment, control, test situation, variable, aim, independent variable, dependent variable, controlled variable, replicated, conclusion.

Equipment list

Each PAIR will require:

- gas jar or straight sided jam jar
- two broad bean seeds
- paper towel
- cheap filler material like potting mix, rice hulls or sand
- marking pen or small label.

Each STUDENT will require:

- **Notebook.**

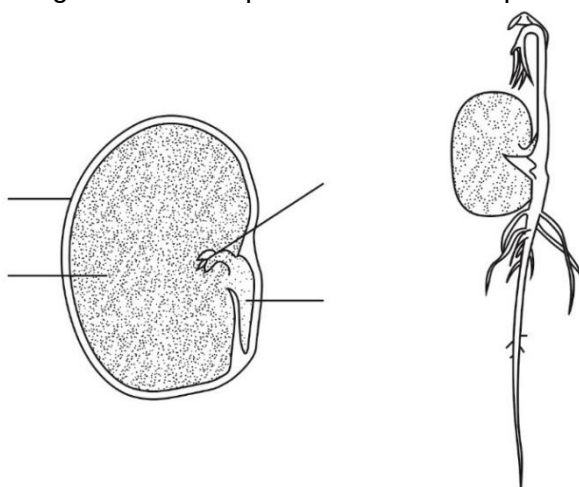


Things to consider:

- **Hints for growing beans:** Gas jars are ideal for this activity because the sides are vertical. Use kitchen paper cut to the height of the jar – a double layer works well. Roll the paper into a tube and insert it into the gas jar. Lightly fill the centre of the paper tube with potting mix or some other absorbent filler. Push a bean seed halfway down the jar, between the paper and the glass. This will enable both developing roots and shoots to be seen clearly and permit measurements as the seedling grows. On the other side of the jar, similarly position the other bean seed. Ensure the paper remains damp by regular watering but that there is never more than a centimetre of water in the bottom. This prevents the bean plant from developing mould.
- Organise a place for the beans to grow that has sunlight during the day but not enough to 'cook' the beans. Peas can also be used. Plan ahead so that you incorporate a weekly lesson during which there is time for watering and observations.
- If you are going to use **Activity 5.9** as an assessment task, ensure you make it clear to students that, although they are collecting data in pairs, they both must **Notebook** as they will be submitting an individual report.
- Make sure students understand a controlled experiment (fair test) and can consider how to make their experiment valid and reliable. They might suggest combining the class data - this would be replication. For the class to share data, you might want to set up a table on a noticeboard or in a class space on your computer network.
- Encourage students to think of quantitative observations e.g. length of shoot, length of radicle, number of lateral roots, as well as general qualitative observations. Tables should be organised at this stage to collect data efficiently. They may also like to draw diagrams and take photographs of the bean plants as they develop.
- Soak some broad beans overnight before this lesson. Use these soaked beans to easily remove the testa and show students the cotyledons, plumule and radicle.
- Potting mix may contain harmful microorganisms. Precautions should be taken if using potting mix: wear gloves, keep potting mix damp, avoid generating or inhaling dust. Wash hands thoroughly after handling.

Teacher content information:

Broad beans, like all legumes, are dicotyledons. This means the seeds have two cotyledons. Each contains starch that is broken down to glucose to provide an energy source for the germinating seedling until it develops leaves to enable photosynthesis to occur.



The cotyledons are covered by a tough skin (testa). The seed contains an embryo consisting of a developing shoot (plumule) and root (radicle).

As you teach:

Step 1: Introduce the lesson by explaining that the students will grow broad bean plants and design an experiment to investigate the characteristics that these living things show over the next few weeks. You may want students to form pairs at this stage and discuss the design of the experiment and the observations they will make. This could be followed with a class discussion to share ideas and review whether their ideas would make a reliable fair test, based as much as possible on quantitative results.

Step 2: Before the students plant their beans, review the structure of a bean seed. If you have soaked spare seeds, the students might be interested in investigating their structure. This is easily done by removing the testa and separating the two cotyledons carefully. One of the cotyledons will contain the embryo and you can point out the plumule and radicle.

Step 3: Students position their broad beans in the jar and place it the selected growing area. In pairs, students design tables to collect their results over at least six weeks.

Suggested question/s:

- What will we need to do to make this a fair test?
- What is a variable?
- Give some examples of variables that might be important to control in this experiment.
- How will we know if the results are reliable?
- What do you think will emerge first, the root or the shoot? Why?
- What observations could you make as the beans grow?
- What measurements could you make?

Step 4: In subsequent lessons students will eagerly wish to monitor their bean seedling's progress. Ensure they take regular measurements and that the seedlings receive water to maintain a 1 cm level in the bottom of the jar.

Follow up:

Some students may want to take their beans home at the end of the unit. If they plant them in a pot and take care of them, students can enjoy the development of flowers and fruit on their bean plants and bring in photographs (or beans) to show to the class the following term.



1.4 Is Sammy alive?

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

ACSHE135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.

ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

Lesson outcomes

At the end of this activity students will be able to:

- discuss the characteristics needed for a person to be considered alive.

What ideas might your students already have?

Students will have varying ideas about how replacing body parts affects being alive and what being alive actually means.

Key vocabulary:

Technology, dialysis, synthetic, intravenous, prosthetic, transplant, intestine, aorta, prosthesis, heart-lung machine.

Equipment list

The CLASS will require:

- internet access

Each STUDENT will require:

- **e-Notebook**

Things to consider:

- Enhance this activity's learning impact by enthusiastically introducing how science and technology have changed our lives (see *Student Guide*), before introducing the animation. It is important that students discuss their points of view freely in small groups, and are encouraged to justify their ideas by linking with the characteristics of life.
- At the second viewing of the animation, stop the video each time *Is Sammy alive?* appears on the screen and discuss each stage in his life.
- A video transcript of *Is Sammy Alive?* may assist some students (see Teacher content information).

Teacher content information

The following transcript exists in many places on the internet and regrettably; the original author is unknown. We have created an animation of this story to make it more engaging for students.



Transcript of animation - Is Sammy alive?

Sammy was a normal, healthy boy. There was nothing in his life to indicate he was any different from anyone else. When he completed high school, he obtained a job in a factory, operating a machine press. On this job he had an accident and lost his hand. It was replaced with an artificial hand that looked and operated almost like a real one.

Is Sammy Alive?

Soon afterward, Sammy developed a severe intestinal difficulty and a large portion of his lower intestine had to be removed. It was replaced with an elastic silicon tube.

Is Sammy Alive?

Everything looked good for Sammy until he was involved in a serious car accident. Both of his legs and his good arm were crushed and had to be amputated. He also lost an ear. Artificial legs enabled Sammy to walk again, and an artificial arm replaced the real arm. Plastic surgery enabled doctors to rebuild the ear.

Is Sammy Alive?

Over the next several years, Sammy was plagued with internal disorders. First, he had to have an operation to remove his aorta and replace it with a synthetic vessel. Next, he developed a kidney malfunction, and the only way he could survive was to use a kidney dialysis machine (no donor was found for a kidney transplant). Later, his digestive system became cancerous and was removed. He received nourishment intravenously. Finally, his heart failed. Luckily for Sammy, a donor heart was available, and he had a heart transplant.

Is Sammy Alive?

It was now obvious Sammy had become a medical phenomenon. He had artificial limbs, nourishment was supplied to him through his veins; therefore he had no solid wastes. Dialysis removed all waste material. The heart that pumped his blood to carry oxygen and food to his cells was not his original heart. But Sammy's transplanted heart began to fail. He was immediately placed on a heart-lung machine. This supplied oxygen and removed carbon dioxide from his blood, and it circulated blood through his body.

Is Sammy Alive?

The doctors consulted bioengineers about Sammy. Because almost all of his life-sustaining functions were being carried on by machine, it might be possible to compress all these machines into one mobile unit, which would be controlled by electrical impulses from Sammy's brain. This unit would be equipped with mechanical arms to enable him to perform manipulative tasks. A mechanism to create a flow of air over his vocal cords might enable him to speak. To do all this, they would have to amputate at the neck and attach his head to the machine, which would then supply all nutrients to his brain. Sammy consented and the operation was successfully performed.

Is Sammy Alive?

Sammy functioned well for a few years. However, a slow deterioration of his brain cells was observed and was diagnosed as terminal. So the medical team that had developed around Sammy began to program his brain. A miniature computer was developed: it could be housed in a machine that was humanlike in appearance, movement, and mannerisms. As the computer was installed, Sammy's brain cells completely deteriorated. Sammy was once again able to leave the hospital with complete assurance that he would not return with biological illness.



Is Sammy Alive?

If Sammy is not alive at the story's end, exactly when did he stop being alive?

Lesson plan

Step 1: Introduce this lesson with a discussion on how science and technology have improved our quality of life and helped save lives, using the stimulus material in the Student Guide. Ask students if they can think of other examples.

Step 2: The class should watch the animation together for an overview.

Step 3: Ask students to access the **Notebook** exercise. They may copy the table or download the **e-Notebook**.

Step 4: Watch the video a second time but stop the video after each stage and let students discuss in small groups what has happened to Sammy and how his life has changed. Students add information to their **Notebook** tables. Remind students to look back at the characteristics of life and pose the question - Is Sammy alive?

Step 5: A spokesperson from each group then reports to the class. Give a short time for comments/discussion after each report. Encourage open-mindedness to differing points of view that are backed up with reasons.

Suggested questions:

- What happened to Sammy?
- How will this affect his life?
- Is he still alive?
- What are the essential part(s) of the body that must work for us to be alive?
- Can you relate this to the characteristics of life?



2.1 Cell theory

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Cells are the basic units of living things.
- Understanding cell structure and function has been possible due to the development of microscopes.
- Living organisms can be unicellular or multicellular.

ACSHE134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.

ACSHE226 Science knowledge can develop through collaboration across the disciplines of science and the contributions of people from a range of cultures.

ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

Lesson outcomes

At the end of this activity students will be able to:

- discuss how the development of microscopes has led to an increased understanding of cells as the basic unit of living things
- describe the different types of microscopes and their capabilities and limitations.

Key vocabulary:

Robert Hooke, cells, Antonie van Leeuwenhoek, Royal Society, microorganisms, Robert Brown, cell theory, compound light microscope, electron beam, transmission electron microscope, scanning electron microscope, magnification, resolution, lens, unicellular, Nucleus.

Equipment list

The GROUP will require:

- internet access

Each STUDENT will require:

- **Notebook**

Teacher content information

Light microscopes use glass lenses to focus light that passes through a specimen. An objective lens near the specimen collects light to form an image about 10 times the size of the specimen. The eyepiece lens magnifies this image (X4, X10, X40), enabling an image up to X400 in a laboratory microscope. The maximum resolution for light microscopes is about 0.2 micrometres and magnification can be up to 2000 times the actual size.

Electron microscopes use magnets and a beam of electrons to form an image. They have resolutions about 1000 times better than that of light microscopes (0.2 nanometres). The electron beams pass through vacuum chambers, meaning specimens are not alive when viewed. This was seen as a problem in early electron microscopes but has been overcome.



There are two types of electron microscope:

- In **transmission electron microscopes (TEM)** a beam of electrons passes through a thin section of a stained specimen in a vacuum chamber. The image formed can be up to 5,000,000 times.
- In **scanning electron microscopes (SEM)** a beam of electrons in a vacuum chamber scans the surface of a sample coated with a thin film of a metal, such as gold. A three-dimensional image of the surface of a specimen can be obtained. Magnification can be up to 500,000 times.

The **Royal Society of London** was inaugurated in 1660 for “improving natural knowledge”. It is possibly the oldest such society in existence. It was granted a royal charter by King Charles II. The society today acts as a scientific advisor to the British Government, funding research fellowships and scientific projects. Robert Hooke was appointed Curator of Experiments in November 1666 and the society’s early meetings included experiments performed by him. The society had many eminent British scientists as members, e.g. Charles Darwin, Isaac Newton, Robert Boyle, John Dalton, Humphrey Davy, Alexander Fleming, Joseph Lister, as well as overseas scientists such as Albert Einstein and Sigmund Freud. The sharing and discussion of ideas at meetings led to rapid advancements in scientific understanding, particularly in the 17th and 18th Centuries.

Lesson plan

Step 1: Introduce this activity by posing questions to see what preconceived ideas students have about the discovery of cells. Pages 12 to 15 of the *Student Guide* could be used as an individual reading activity. You could then project the pages on a screen and discuss the development of microscopes with the help of the discussion questions on page 15, or have students do this in groups.

Step 2: Bring the class together to compare answers to the questions and then look at page 15. Allow students access to the digital interactive to compare the light microscopes with the electron microscopes. When students plot the historical developments on a scaled timeline they should appreciate the flurry of improvements made in the last 150 years.

Step 3: Direct students to the **Notebook** tasks.

Step 4: Bring the class back together to review answers to questions and get students to try to summarise what they have learnt about the links between advances in technology and discoveries in science.

Step 5: Ask students to explore the *Find out more* section and discuss the amazing scanning electron microscope (SEM) images taken by Dr Roger Heady of the ANU.

Suggested questions:

- When (which century) do you think people first realised that living things were made of cells?
- What made this discovery possible?
- What part of the light microscope most limits the magnification of the image?
- Why was the cell theory so important?
- What role did the Royal Society play in the advancement of science?
- How are electron microscopes able to obtain images with such high magnification?
- Can you think of any disadvantages that electron microscopes have?
- What is the main difference between SEM and TEM?



2.2 How small can things be?

Curriculum links

- ACSYS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- ACSYS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- appreciate the vast differences in the size of objects
- use appropriate units when measuring macroscopic and microscopic objects
- draw and interpret scale diagrams.

What ideas might your students already have?

Students probably will not have considered the units used for measuring microscopic objects and will not be familiar with μm or nm . They may believe that atoms can be seen through a light microscope, as they cannot appreciate the huge difference in orders of magnitude between the size of a small macroscopic object and atomic particles.

Key vocabulary:

Scale, micrometre (μm), nanometre (nm), picometre (pm).

Equipment list

The CLASS will require:

- internet access

Each GROUP will require:

- ruler
- post-it pad
- small nail
- coin (10c or 20c)
- watch glass /petri dish.

Each STUDENT will require:

- **Notebook**

Things to consider:

- Some students will have difficulty with the mathematical and spacial concepts in this activity. It may be helpful to group students with good mathematical ability with weaker students, or group weaker students together and give them more structured help.
- Familiarise yourself with the animation *Cell size and scale*. This enables students to zoom in from a macroscopic object (a coffee bean) through all the objects in their list to the carbon atom. They see what is visible at each magnification and the relative sizes of the objects, with a changing scale of appropriate units in the corner of the screen.



Teacher content information

Size: the smallest objects the unaided human eye can see are about 0.1 mm long. That means that under the right conditions, you might be able to see an amoeba, a human egg, and a paramecium without using magnification. A magnifying glass can help you to see them more clearly, but they will be tiny.

Smaller cells are easily visible under a light microscope. It's even possible to make out structures within the cell, such as the nucleus, mitochondria and chloroplasts. Light microscopes use a system of lenses to magnify an image. The power of a light microscope is limited by the wavelength of visible light, which is about 500 nm. The most powerful light microscopes can resolve bacteria but not viruses.

To see anything smaller than 500 nm, you will need an electron microscope. Electron microscopes shoot a high-voltage beam of electrons onto or through an object, which deflects and absorbs some of the electrons. Resolution is still limited by the wavelength of the electron beam, but this wavelength is much smaller than that of visible light. The most powerful electron microscopes can resolve molecules and even individual atoms.

Lesson plan

Step 1: Introduce the lesson by asking students for the largest and smallest living things they can think of and asking them to estimate their sizes.

Step 2: Direct students to the task of measuring and calculating the thickness of a post-it note. Let students use their own units before using the discussion questions to introduce the idea of appropriateness of mm and μm in this context. Then explain the conversion to these units from units they are familiar with, e.g. km, m and mm.

Step 3: Introduce the idea of scale diagrams and ask for everyday suggestions for the use of them, e.g. maps, illustrations in books. Students will then practice drawing scale diagrams, using the nail and coin. Stop at a convenient point to discuss the calculation and compare students' answers. You may want to complete the discussion questions as a larger group activity, if students have problems with the conversions.

Step 4: Scale diagrams of three different egg cells are presented on page 19. Discuss the information on this page.

Step 5: Ask students to predict the order of size of the objects on page 20 and tabulate their results. Ask students to choose appropriate measurement units for each object. Discuss students' responses briefly and allow them to modify their table after discussion with others, but do not yet give correct answers.

Step 6: Students now explore the animation *Cell size and scale* in the digital link and complete columns 3 and 4 of their table, followed by the discussion questions. After they have completed this activity, conclude with a discussion about what they got correct and what misconceptions they had about the size of objects.

Suggested questions:

- What does the word scale mean to you?
- Can it have any other meaning?
- How much bigger is a kilometre than a metre (a metre than a millimetre etc.)?
- What units would you use to measure ...?
- What units would you use to measure a cell (molecule of water)?
- Can you think of everyday examples of uses for scale diagrams or drawings?



2.3 Making little things bigger

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Understanding cell structure and function has been possible due to the development of microscopes.

ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- recall the parts of a light microscope and their functions
- focus a microscope under low and high power to view prepared slides.

What ideas might your students already have?

If students have used a microscope before, they may not have learnt correctly or do not accurately remember procedures. It is important to take time to ensure correct techniques are learnt.

Key vocabulary:

Objective lens, eyepiece, iris diaphragm, condenser, coarse focus, fine focus, stage, slide clips, parfocal, field of view, magnification.

Equipment list

The CLASS will require:

- internet access

Each PAIR will require:

- compound light microscope
- lens tissues
- prepared microscope slides such as:-
 - slide of insect mouthparts
 - slide of plant stem cross section (TS)

Things to consider:

- Check that microscopes are clean and correctly adjusted before the lesson as students will be discouraged if they follow instructions but still cannot observe the specimens clearly.
- Explain that the microscope is parfocal. This means there is no need to adjust the coarse focus when changing to a higher magnification if the slide has been correctly focused under low power. Only minor adjustments using the fine focus knob may be needed, together with adjustment of light.
- Would your class be able to follow the instructions in the Student Guide on the correct use of the microscope? Would a teacher demonstration be beneficial?



- Continually reinforce the correct procedures for microscope use, so that students do not get into bad habits. Common problems that should be corrected are:
 - Students should not use the fine focus knob when under low power (X4 or X10 objective) as this takes the knob to one end of its range and then it is unable to be used for fine control under high power.
 - Students should never use the coarse focus knob under high power (x40).
 - Students should adjust the light intensity using the iris diaphragm when they move from low to high power, to optimise the image.

Teacher content information

Correct procedures for use of a microscope

1. Check that the eyepiece lens and objective lens are clean by looking down the barrel. Check the mirror is clean. Wipe with lens tissue and alcohol if necessary.
2. The condenser lens concentrates the light that is focused through the object and is situated (if present) under the stage of a microscope. Condensers should be correctly adjusted prior to the lesson. If so instruct students not to touch the condenser knob. If not adjusted, ask students to raise the condenser as high as possible.
3. Rotate the nosepiece to select the low power lens (X10) suggested when viewing cells.
4. Adjust the angle of the lamp and mirror so that light is directed up the barrel.
5. Place the slide on the stage and secure with clips. (If using a wet mount make sure the slide is dry and has a cover slip.)
6. Adjust the light intensity using the iris diaphragm.
7. Looking from the side, lower the objective lens until it is just above the slide.
8. Look through the eyepiece and use the coarse focus knob to move the objective lens up until the object is in focus.
9. To draw cells or view the specimen under high power (X400), carefully rotate the nosepiece until the X40 lens is in position. The lens will fit if correctly focused under low power. The object should be visible, as most microscopes are parfocal. Adjust the image using the fine focus knob only.

NB: Do not touch the coarse focus knob again, once you have focused under low power.

Lesson plan

- Step 1:** Begin by explaining that students will do much microscopy work in this unit. Stress that this can be very exciting if they learn the parts of the microscope, the correct procedures and take good care of the microscopes.
- Step 2:** Direct students to undertake the microscope parts digital activity. Give them five to ten minutes to complete and then ask them to do the quiz.
- Step 3:** When they complete the quiz, they are ready to use their microscopes. Students are issued with microscopes (in pairs). Demonstrate the correct way to carry a microscope is to hold it by the arm with one hand and by the base with the other. Emphasise they should not touch lenses or fiddle with knobs, until they know how to use them.



The instructions are on pages 21 and 22 of the *Student Guide*. Circulate constantly to ensure all students follow the guidelines in the correct order. Stress the need to check this sequence of procedures every time they focus a slide. You might want to stop the class at appropriate times to discuss the questions and solve common problems.

Step 4: Students should then observe the prepared slides, comparing what they can see under low/high power, and experimenting with lighting using the iris diaphragm.

Suggested questions:

- What is the difference between a macroscopic and microscopic organism? Can you give examples?
- Can you explain what parfocal means?
- What magnification would you get with a X4 objective lens and a X10 (etc.) eyepiece?
- Why is an image of a specimen sometimes not as clear when viewed under a higher magnification?
- If you wanted to move a slide to look at cells above and to the right of those you are observing, how would you move it?
- Describe how adjusting the iris diaphragm affects what you can see through the microscope?



2.4 Observing cells

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Cells are the basic units of living things.
- Differences in cell structure can be used to classify living things.
- Understanding cell structure and function has been possible due to the development of microscopes.

ACSHE134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.

Lesson outcomes

At the end of this activity students will be able to:

- identify cell components using a light microscope.

What ideas might your students already have?

If students have used a microscope before, they may not have learnt correctly or do not accurately remember procedures. It is important to take time to ensure correct techniques are learnt.

Key vocabulary:

Cell wall, cell membrane, nucleus, cytoplasm, chloroplast.

Equipment list

Each STUDENT will require:

- **Notebook**

Things to consider:

Familiarise yourself with the virtual microscope link '*You are looking at the microscope*'. Students will need to turn on the audio to hear the tour, or use closed captions. Suggest that students bring earphones.



Teacher content information

Comparing structures in cells

Cell parts visible (✓ or X)	Human cheek cell	Elodea leaf cell (water plant)	Onion skin cell
nucleus	✓	X (not visible but present)	✓
cytoplasm	✓	✓	✓
cell wall	X	✓	✓
cell membrane	✓	✓	✓
chloroplasts	X	✓	X

As you teach:

Step 1: To start, have the students study the diagram of a typical plant cell and learn the names and functions of the parts they might see using their school microscopes.

You might like to do an overview with the whole class, using the image on a screen, then give students 10 minutes to learn the information with a partner.

Step 2: Page 24 in the **Student Guide** shows labelled micrographs of cells, which students examine to locate the parts present. They then complete a table to compare the structures found in each type of cell.

Suggested questions:

- How can you tell whether a cell has a cell wall, as well as a cell membrane?
- How would you distinguish between a nucleus and a chloroplast?



2.5 Microscope skills

Curriculum links

AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- describe that an image viewed with a microscope is inverted and reversed
- explain why resolution can be a limiting factor when viewing magnified objects
- use a minigrid to measure the size of objects under a microscope.

Key vocabulary:

Resolution, minigrid, diameter, field of view.

Equipment list

Each PAIR will require:

- microscope
- microscope slide
- sticky tape
- newsprint
- coloured magazine page
- scissors
- minigrid
- selection of prepared slides of cells.

Each STUDENT will require:

- **Notebook**

Things to consider:

- Continue to reinforce the procedures for microscope use by reviewing the sequence of steps at the start of the lesson, including safe carrying and ensuring the microscope is clean.
- Use prepared slides with clear, even-sized cells, to make counting and measurement more straightforward. Encourage students to think for themselves how to find the length of one cell, if they know the diameter of the field of view.
- If you do not have commercially prepared minigrids on slides, they can be made by photocopying a sheet of mm graph paper onto an overhead transparency and cutting into cm squares.

Teacher content information

When objects are viewed under a light microscope, the light refracting through the two lenses makes the object appear upside down and inverted sideways. This means that the slide must be moved opposite to what seems to be the correct direction, as seen looking down the eyepiece.



Resolution is the ability to separate clearly small details in an object. As an object is viewed under higher magnification, the image becomes larger, but the object itself can become less clear, so that the detail becomes blurred. This is because points that were close together become separated and can be seen as individual dots, rather than merging into one clear image.

Lesson plan

Step 1: Introduce the lesson by explaining to the class that they will be using their microscope to learn more about how they work, including how to find out the size of objects viewed. Review, with student help, the sequence of steps that must be carried out to use a microscope correctly and to keep it in good condition.

Part A

Step 2: Have students set up their microscopes and work through the activities on page 25. Ensure that they cut small pieces of newsprint (1 cm²) so that the light is not blocked too much. When the class has explored the letter 'e' and the coloured picture, go through the discussion questions to ensure the concepts of inversion and resolution are understood.

Part B

Step 3: Students may now work out the diameter of the field of view of their microscopes, starting at low magnification (X4 objective lens if present) and then at higher magnification. They should draw up the table and convert their measurements to μm . Explain that once they know the diameter of the field of view of their microscope, they can use this information to find the size of other objects viewed.

Step 4: Students now view prepared slides of cells and use the information from the minigrd to work out the length of cells. Discuss with students how they can find the length of one cell. Finish the lesson by comparing measurements obtained by students for the cells used. Make sure you leave time to pack the microscopes up carefully.

Suggested questions:

- Describe how the letter 'e' appeared under the microscope.
- Why do you think this happens?
- What differences can you see in the image of the 'e'/coloured picture when you increase the magnification?
- Why is the image less clear?
- How can you convert mm into μm ?
- How can we find the length of a cell if we know the diameter of the field of view?



2.6 Looking at living cells

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Differences in cell structure can be used to classify living things.
- Understanding cell structure and function has been possible due to the development of microscopes.

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- prepare a wet mount of cells
- add a stain to a wet mount to show up various cell components.

Key vocabulary:

Wet mount, pipette, methylene blue stain, iodine solution, artefacts.

Equipment list

The CLASS will require:

- internet access

Each PAIR will require:

- piece of onion skin (the transparent film between the onion layers)
- 2 microscope slides
- 2 cover slips
- eye dropper/pipette
- water
- dropper of methylene blue stain
- dropper of iodine solution
- probe/toothpick, scissors and scalpel
- tissues.

Each STUDENT will require:

- **Notebook**



Things to consider:

- Reinforce correct microscope procedures. It is particularly important that students are aware of the importance of drying the slide, including underneath, before placing it on the stage of the microscope. Moisture can damage the lenses.

Tips:

- Cut segments of onion about 1 cm wide. From one leaf of the segment, peel off the skin by pulling at an angle from the flesh. This skin can be placed on a slide. Ensure there are no folds.
- Add a drop of water to the slide so that, when the coverslip is lowered at an angle, there is sufficient water to reach the edges of the coverslip, but not enough to make the whole slide wet. If the drop is not big enough, or the slide is not lowered gradually, there will be air bubbles under the coverslip.
- To stain the cells, place a drop of stain at one edge of the coverslip and draw the stain through from the opposite edge with a tissue. To help the stain to spread out put a drop of water in the same position at the edge of the coverslip and draw it through with the tissue. The cells should now be evenly stained.

Teacher content information

Effects of stains:

- **Methylene blue** stains the nucleus blue.
- **Iodine** enhances the imaging of the shape and parts of the cell, especially when cells are quite clear in colour. The cell wall and membrane, as well as the organelles, have more contrast. If the cells have starch in them as a long-term storage product, that part will turn blue-black.

Stains react with specific chemicals in cells and so can be used to pick out selected tissues or organelles clearly. However, sometimes stains can mask other things in the cells.

Chloroplasts will not be present in the cells of the onion bulb because it grows below ground. The chloroplasts would be present if you took leafy material (above ground) from the onion plant.

Lesson plan

- Step 1:** Introduce the lesson by informing students they are going to make their own slides. Show the class the video *How to make a wet mount*, stopping to clarify points as you go. If students have laptops in the laboratory they could view this again individually. You may also want to do a demonstration before the students begin. Staining can be demonstrated at a later stage, after the students have first made a wet mount. Make sure you mention prevention of artefacts (point out the illustrations on page 27) and the importance of not putting wet slides on the microscope stage.
- Step 2:** Reinforce the microscope procedures before students set up their microscopes and follow the video to make a wet mount. They then follow instructions (pages 21 – 22) to focus on low and high power.
- Step 3:** Students then make another wet mount. This time they stain with methylene blue and comment on the difference this makes to what they can see. Quicker students may like to make a third slide using iodine stain and note the differences in the staining from methylene blue.



Step 4: Discuss the difference that staining makes to the image viewed. Instruct students to pack away their microscopes carefully, ensuring the stage is dry.

Suggested questions:

- How are the stained cells different from those mounted only in water?
- What part(s) of the cell are stained by methylene blue/iodine?
- What advantages does staining have when preparing wet mounts of cells?
- Can you think of any disadvantages of staining?

Follow up: More wet mounts will be made in **Activities 3.3, 4.1 and 4.2.**



△ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

2.7 Drawing cells

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Differences in cell structure can be used to classify living things.
- Understanding cell structure and function has been possible due to the development of microscopes.

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- make a labelled biological drawing that is accurate, to scale and in proportion.

What ideas might your students already have?

Students have a tendency to draw a lot of cells, too small, instead of one or two large cells. They also sketch and shade their diagrams as they would in Visual Arts.

Equipment list

The CLASS will require:

- internet access

Each STUDENT will require:

- ruler
- sharp HB pencil
- eraser
- **Notebook.**

Things to consider:

- A sample set of drawing rules are provided as a guide in the Teacher Content Information below. However, it is better to help the class develop its own set of rules so students have greater ownership and can readily remember them. Develop a class set of rules that includes most aspects shown below.
- Students learn biological drawing more easily if they practice drawing from photomicrographs before attempting to draw cells observed directly with a microscope. Circulate frequently to check student drawings. Do not accept first efforts if they are poor. Some students find it hard to get into the habit of drawing single lines. Positive reinforcement for improvement is better than accepting drawings which do not follow the class rules.
- The key points to stress in biological diagrams are:

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▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

- Draw one or two large cells (minimum length 5 cm for an onion cell).
 - Lines should be drawn freehand but not sketched. A sharp HB pencil is best.
 - No shading or colouring.
 - Only draw what you can see, e.g. if a nucleus is not visible do not draw it, even if you think it should be there.
 - Labels are ruled lines with no arrowheads, touching the part being labelled.
- Students keeping an **e-Notebook** may like to scan or photograph their pencil drawings for addition to their **e-Notebook**.
- You might like to set for homework a reading/drawing/interpretation exercise on jellyfish nematocysts. Download the pdf Specialised stinging cells available from the Queensland Museum site (<http://www.qm.qld.gov.au/~media/Documents/Learning+resources/QMSB/Teacher+and+student+resources/specialised-stinging-cells.pdf>)

Teacher content information

Biological Drawing Rules (suggested)

Aim: To draw a diagram that is accurate, to scale and in proportion.

All diagrams should:

- be drawn and labelled using a sharp HB pencil
- be two dimensional
- be large and clear
- include a scale and heading
- have lines for labels drawn with a ruler and no arrowheads
- have label lines horizontal if possible and not crossing over each other.

Ensure you:

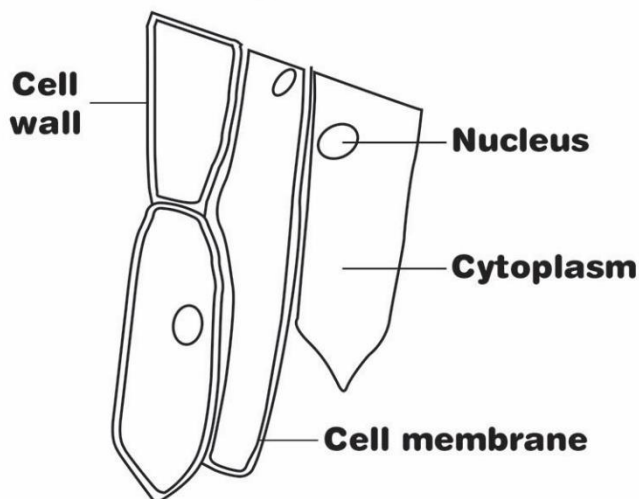
- draw clear, smooth, continuous lines
- draw structures you can see clearly
- indicate cut sections with the words cross section in the heading.



DRAWING FROM A MICROSCOPE

Onion skin

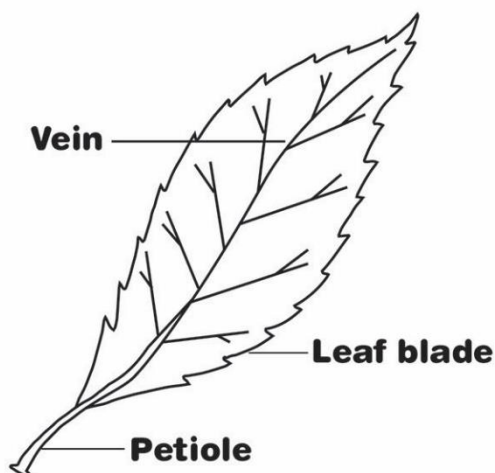
Microscopic magnification x400



SCALE DRAWING OF A LARGER SPECIMEN

Leaf

Magnification x2



DO NOT
SHADE

DO NOT USE A RULER FOR DRAWING

DO NOT
SKETCH

Lesson plan

- Step 1:** Introduce the lesson by discussing how scientists record what they view under the microscope (remembering that it is only relatively recently that they have been able to take photographs and make digital images). Examples of biological drawing are shown in the *Student Guide* page 28.
- Step 2:** Ask students to download the PowerPoint of ten drawings made by Year 8 students from the *Student Guide*. Organise students to work in small groups and discuss which drawings show the details best and in the most scientific way (accurate, to scale and in proportion).
- Step 3:** Encourage a class discussion and develop a class set of rules for biological drawings.
- Step 4:** Ask students to pick their own cell in the onion cell micrograph on page 29 and to draw it using the class set of drawing rules. They then ask a fellow student to check the rules have been followed and to guess which cell has been drawn.
- Step 5:** Have students practice their drawing skills using any of the photomicrographs on subsequent pages of the *Student Guide*. Circulate to check the diagrams, showing some of the best ones to the class.



Suggested questions:

- What is important to show in a biological drawing?
- Which diagrams show the cells most clearly and scientifically?
- What do these diagrams have in common that might help us work out some rules for the class?
- Can you tell which cell your partner has drawn? Why/why not?
- Have you checked that your diagram follows all the class rules?

Follow up:

- There is opportunity for further biological diagrams in **Activity 4.1**.
- Explore the Queensland Museum activity - *Specialised stinging cells* (refer to *Things to consider*).



3.1 What's inside a cell?

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Differences in cell structure can be used to classify living things.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- classify cells according to their shape and structure as plant, animal, bacterial or fungal.

Key vocabulary:

Fungus, bacteria, classification, vacuole, nucleus, cell membrane, cell wall, cytoplasm, food storage granule, mitochondria, chloroplast, chromosome, cellulose, chitin, murein, turgor.

Equipment list

Each STUDENT will require:

- **Notebook.**

Things to consider:

It is important to point out all the features of cells you want students to look for and to stress that a combination of characteristics is often needed to be sure of classification. Many students may think that all plant cells have chloroplasts, so point out some that do not, e.g. leaf epidermis, stem and root cells.

Ensure students understand the colour of prepared slides is often artificial and due to stains highlighting specific cell types and organelles.

Teacher content information

Early classification systems of living things were based on general distinguishing features and scientists tried to categorise all organisms as plants or animals. However, with more detailed study of cell structure, particularly with the use of electron microscopy, it soon became clear that cell differences were an essential part of classification.

- **Plant cells** are surrounded by a rigid cellulose cell wall and have a regular, often box-like, shape. Cells have nuclei and can have large vacuoles containing cell sap. Cells in leaves or other photosynthetic parts contain chloroplasts; however, it is important that students understand that not all plant cells have chloroplasts, so that the presence of chloroplasts needs to be looked at, together with other cell features, for classification purposes.
- **Animal cells** have a nucleus but no cell wall and only small vacuoles (if present). Cells are surrounded by a membrane and are often irregular in shape.



- **Fungal cells are box-like, with nuclei and cell walls, and so are similar to plant cells.** Significant differences are that cell walls are made of chitin (the same material comprising exoskeletons of arthropods like insects and prawns), not cellulose. Chloroplasts are never present. Fungi are thus heterotrophic and placed in a separate Kingdom, the Fungi.
- **Bacterial cells:** These differ from other cells because they have no nucleus (prokaryotic). The nuclear material is free in the cytoplasm (as a circular chromosome) and is not visible under a light microscope. They are surrounded by a cell wall made of murein (peptidoglycan), a sugar and amino acid polymer. There are no cellular organelles (such as mitochondria, chloroplasts) present. Bacteria are therefore placed in a separate Kingdom, the Monera.

Lesson plan

- Step 1:** Begin the lesson by explaining that biologists can use cell structure to determine what type of organism they are studying; i.e. cells can be used to classify living things. Divide the class into small groups to discuss the stylised diagrams on page 35 in the *Student Guide*. Bring the class back together to decide what characteristics you would use to distinguish cells as belonging to each group of organisms.
- Step 2:** Project the digital activity onto the class screen and show the short video illustrating the three main differences between plant and animal cells.
- Step 3:** Ask students to complete the **Notebook** tasks and classify the cells A – D.
- Step 4:** Summarise the lesson by discussing the way that groups of organisms can be classified by cell structure.

Suggested questions:

- Are there any characteristics that all these cells have?
- What differences can you see?
- Can we classify any group on one characteristic only?
- What combinations of characteristics are useful to separate the cell types?



3.2 What makes cells function?

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Understanding cell structure and function has been possible due to the development of microscopes.

Lesson outcomes

At the end of this activity students will be able to:

- discuss the roles of organelles in cell metabolism
- develop literacy by exploring and reading digital texts to comprehend and communicate scientific information.

Key vocabulary:

Organelle, mitochondrion (mitochondria), chloroplast, nucleus, cell membrane, metabolism, Paramecium, Euglena, Amoeba, oral groove, cilia, chloroplasts, flagellum, eyespot, pseudopod, phagocytosis.

Equipment list

The **CLASS** will require:

- internet access

Each **STUDENT** will require:

- **Notebook**

Things to consider:

Jigsaw is a strategy for group work that allows students to cover a broader amount of information in a shorter time. See *Science By Doing* Professional Learning Resource *Inquiry-based Teaching* page 11.

- Think about the structure of your groups beforehand. Spreading weaker students between groups is probably a good idea.
- Familiarise yourself with the '*biology4 kids*' website that this activity links to. It has clearly written information and diagrams about each organelle.
- The 360° video *Virtual Plant Cell: Cell Explore* is available for viewing on: YouTube using a smart phone with or without a cardboard headset, tablet/iPad or a computer.
- To view on a smart phone with cardboard headset (immersive viewing) you need: a smart phone, the YouTube app, a cardboard headset (<https://vr.google.com/cardboard/get-cardboard/>) and internet connection.
 - Open the YouTube app
 - Search for Plant Energy Biology
 - Select Virtual Plant Cell: Cell Explore (4.54 min)
 - Tap the cardboard icon (bottom right)
 - Insert smart phone into cardboard headset for 360° viewing.



- To view on smart phone or tablet/iPad
 - Open the YouTube app
 - Search for Plant Energy Biology
 - Select Virtual Plant Cell: Cell Explore (4.54 min)
 - Move device for 360° viewing

Lesson plan

Step 1: Explain the idea of a **Jigsaw** activity and discuss the task, referring to the organelle diagrams in the *Student Guide* page 36. Divide students into groups, giving them time to decide on roles and ask questions about the task.

Step 2: Students have time to individually research their organelle. Circulate to ensure all students are making progress and to estimate the best time to reform groups (20 minutes is probably enough).

Step 3: Students report back to their groups and follow the instructions in the *Student Guide*.

Step 4: Group representatives report back to the class. Make sure you give opportunities for students to compare and discuss findings and extend their understanding of organelles and their roles in effective cell metabolism.

Step 5: The **Notebook** activity introduces several common single-celled organisms and invites students to apply their understanding of cell organelles to these new cell types.

Suggested questions:

- What do you think the word *organelle* means?
- What do all organelles have in common?
- Why are cell membranes so important?
- Which organelle is involved in producing energy for the cell?
- Which organelle manufactures food in some cells?



3.3 Yeast metabolism

Curriculum links

- ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.
KEY IDEAS
- Living organisms can be unicellular or multicellular.
- ACSIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identify improvements.
- ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe some of the characteristics of life shown by yeast cells
- plan and carry out a controlled experiment
- draw and interpret a line graph from experimental data.

What ideas might your students already have?

Students should be able to recall the characteristics of life from activities in Part 1 of this topic. They should also have some understanding of experimental design, which may need reinforcing.

Key vocabulary:

Yeast, budding, dependent variable, independent variable, controlled experiment.

Equipment list

Students are planning their own experiment so requirements will vary.



Suggested equipment per GROUP is:

Part A: What a gas!

- 3 conical flasks (250 mL)
- 3 beakers (250 mL)
- 1 packet of dried yeast
- 3 tsp sugar
- hot tap water
- thermometer
- 3 round balloons
- stirring rod
- marking pen
- measuring cylinder (100 or 250 mL)
- electronic or beam balance.

Part B: Budding cells

- yeast solution (from Part A or previously made)
- microscope slide and coverslip
- eye dropper/pipette
- tissue
- mounted needle.

Each STUDENT will require:

- **Notebook**
- graph paper.

Things to consider:

- Discuss the best way to mix water to get the correct temperature.
- It is important to maintain the temperatures of the flasks during the experiment. A water bath is a good idea. If you have a large trough, these can be maintained by students at the appropriate temperature by adding hot water.
- Discuss control of important variables, measuring being the most accurate. Note the teaspoon measurement in the equipment list and discuss how this could be made more accurate.
- Testing for carbon dioxide - This can be done as a demonstration, using bubbling yeast in a flask with a delivery tube inserted into a test tube of limewater. Students can also try the test by attaching a balloon containing the gas to a delivery tube inserted into a test tube of limewater.
- The amount of gas in the balloons can be compared by using string to measure the balloon diameter. This measurement can be used directly, or more able students may like to calculate the volume of gas (sphere volume is $\frac{4}{3} \pi r^3$). Alternatively, the Archimedes method of pushing the balloon into a full beaker of water and measuring the volume of water displaced can be used.



- There are many demonstration available of the yeast/ balloon set-up
e.g.: <http://www.youtube.com/watch?v=ys47cu4AXPA>
- You may want to do the *Constructing graphs* section of **Activity 6.1** (page 73 *Student Guide*) before students do this experiment to revise drawing line graphs.
- Yeast budding - Stress to students that they will see cells best if they put a tiny drop of yeast onto the slide to ensure that the cells are not too crowded together. You may want to demonstrate or invite students to view slides from other groups who have successfully seen cells. You may want to use a separate flask of yeast solution, prepared ahead in warm conditions, so that it is already active and frothy.
- **Characteristics of life** that the students might observe are: organisation (cells), obtaining and using energy, growth and reproduction. They might also infer that because the reproduction is a part of the cell breaking off, that they are passing on hereditary material, and that the change in rate of gas production indicates a response to a stimulus (temperature change).

Teacher content information:

Yeasts are eukaryotic microorganisms which are classified in the Fungi kingdom. Bakers' yeast is *Saccharomyces cerevisiae*. The reaction of respiration converts sugars into ethanol and carbon dioxide. This reaction is used to make bread rise and also in the production of alcoholic beverages. Reproduction is by budding, a method of asexual reproduction in which a small part of the cell is divided off a parent cell (by mitosis) and then grows to become a new individual.

Other species of yeast, e.g. *Candida albicans*, are opportunistic pathogens and can cause infections, such as tinea (athlete's foot and ringworm) in humans. Yeasts have recently been used to generate electricity using microbial fuel cells and to produce ethanol as an alternative fuel.

Yeast metabolism - Carbon dioxide gas is produced by respiration, using sugar in the solution catalysed by enzymes. The optimal temperature for yeast growth is 30-35° C. Gas production will be slow at 20° C, due to slower chemical activity, and slow down again approaching 60° C, as the enzyme is denatured.

Within the limits of equipment available, students should design their own experiments. A possible design is:

- Step 1:** Divide the yeast equally between the three flasks. Label them 20° C, 40° C and 60° C.
- Step 2:** Add 1 tsp. sugar to each flask. Perhaps weigh equal amounts?
- Step 3:** Half fill each of the beakers with water, and adjust to the right temperature.
- Step 4:** Use a measuring cylinder to add exactly 100 mL water at the correct temperature, stir the mixture, and quickly seal each flask with a balloon.
- Step 5:** Place a flask into each of the three 'water bath' beakers.
- Step 6:** Observe the flask contents and balloons over the next 30 minutes.



Lesson plan

- Step 1:** Begin the lesson with a brainstorm revision of the characteristics of life. Then explain students will investigate yeast to look for signs of these characteristics. A brief discussion about the features of yeast might be appropriate here.
- Step 2:** In Part A, students will design their own controlled experiments. Discuss what factors need to be considered if this is to be a fair test (controlled experiment) and how to control variables accurately. Ensure students understand the terms *independent*, *dependent* and *controlled variable* and can relate these terms to this experiment. Form groups of four and give time to plan each student's role in the group.
- Step 3:** Students set up their experiment. You will need to circulate to help solve any problems with design, equipment use etc. Ensure they understand the importance of maintaining the flasks at the set temperatures.
- Step 4:** It will take at least 30 minutes (leave as long as your lesson will allow) to collect the gas. Use this time to reinforce graphing skills (**Activity 6.1**) and/or set up results tables in **Notebooks**.
- Step 5:** Explain Part B. Students can set up their microscopes when ready to view yeast budding (or set up a demonstration microscope if time is short). Refer students to the images in the **Student Guide** if they are not sure what to look for.
- Step 6:** Discuss the observations that have been obtained and relate these to the characteristics of life. Each group could make a list and then contribute their ideas to a class discussion.
- Step 7:** Students can now draw a line graph for their experiment, with help as necessary in determining axes and scales. They can draw conclusions as indicated in the discussion questions in the *Student Guide*.

Suggested questions:

- What are you controlling in this experiment to make it a fair test?
- How can you control variables accurately?
- Which is the independent variable (the one that you change)?
- Which is the dependent variable (the one that you are measuring)?
- How can we measure the differences in the amount of gas in the balloons?
- How will you keep the temperature constant in each flask?
- What characteristics of life have you observed?



3.4 Cell division

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Living organisms can be unicellular or multicellular.

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Sexual reproduction involves fusion of gametes and cell division, whereas asexual reproduction involves mitosis in a single organism.

Lesson outcomes

At the end of this activity students will be able to:

- describe mitosis as a type of cell division producing two genetically identical cells
- list situations where binary fission and mitosis occur.

What ideas might your students already have?

Students may find it difficult to understand the concept of surface-area: volume ratio. They will probably link increase in size to increase in actual surface area, rather than the inverse relationship when expressed as a ratio.

Key vocabulary:

Binary fission, mitosis, chromosome, surface-area: volume ratio, Paramecium, Euglena, yeast, chromosome, centriole, spindle, spindle fibre, nuclear membrane.

Equipment list

- **The CLASS will require:**
- internet access

Each STUDENT will require:

- **Notebook.**

Things to consider:

- A good introductory discussion is important to make this lesson relevant.
- Cartoon strips should be simple. Do not try to include all the stages in the duplication of chromosomes at this level, just the concept that the chromosomes are copied and each new cell is genetically identical. This can be a short activity, unless you want students to use a computer program (e.g. *Comic Strip Creator*) that can be used to design cartoons. You may want students to draw their cartoons on loose paper to display in the classroom.

Teacher content information:

Unicellular organisms cannot metabolise effectively if they grow past a critical size, at which point their surface-area: volume ratio becomes too small for them to efficiently take in and eliminate substances through the cell membrane. When they reach this critical size the organism divides by binary fission/mitosis (Note – eukaryotic cells reproduce asexually by mitosis and prokaryotic cells by binary fission).



The main reasons for mitosis/binary fission are:

- to reproduce asexually
- to maintain efficient metabolism
- to grow (multicellular organisms).

Lesson plan

Step 1: Begin the lesson with a class discussion with page 40 of the *Student Guide* projected on a screen. Discuss the concept that unicellular organisms carry out all life functions within a single cell and divide into two to reproduce. Use the images to discuss how it is possible, by cell division, that colonies of cells and then multicellular organisms could have developed.

Step 2: Students view the animation of mitosis and test their memory of the process with the click and drag activity. In the **Notebook** activity students consolidate their understanding by drawing a cartoon strip to illustrate the process.

Step 3: Students then view three short videos of cell division in unicellular organisms to compare the cells and processes that occur. Remind the class about the similarities and differences in cells that they learnt in **Activity 3.1**.

Suggested questions:

- Why do you think a unicellular organism would divide in two?
- Why are the two cells formed in mitosis identical?
- How do you think a multicellular organism could develop from one that is unicellular?

Follow up:

Students will develop their understanding of surface-area: volume ratio in **Activity 3.5** and learn more about the role of cell division in reproduction in **Part 5** of this unit.



△ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

3.5 Why are cells so small?

Curriculum links

- ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.
KEY IDEAS
- Cells are the basic units of living things.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- describe the relationship between size of cells and their surface-area: volume ratio
- explain that cells are limited in size because diffusion becomes less efficient as size increases.

What ideas might your students already have?

Students have been introduced to the idea of surface-area: volume ratio decreasing with size in the previous activity, but may still be unsure of this difficult concept.

Key vocabulary:

Binary fission, mitosis, chromosome, surface-area: volume ratio, Paramecium, Euglena, yeast, chromosome, centriole, spindle, spindle fibre, nuclear membrane.

Equipment list

- Each GROUP will require:**
- pink agar cubes with sides of 1 cm, 2 cm and 3 cm (see **Things To Consider**)
- 0.1 M sulfuric acid (100 mL) (see **Things To Consider**)
- 250 ml beaker
- stop watch
- knife or scalpel
- ruler
- paper towel

Each STUDENT will require:

- safety goggles
- Notebook.**

Things to consider:

- Making agar jelly - To make up 500 mL of jelly for the cubes:
 - Weigh out 9 g agar and 2 g solid NaOH.
 - Dissolve the NaOH in 50 mL water.



- Heat 400 mL water in beaker until nearly boiling, Add the agar and stir until dissolved.
- Cool. When the temperature falls below 60⁰ C, add the NaOH solution, stirring continuously.
- Add phenolphthalein solution drop by drop, stirring continuously until the solution is deep pink.
- Pour into a tray so that the depth is 3 cm (this avoids cutting for the larger blocks) and allow to set.

0.1 M H_2SO_4 is 10 mL concentrated sulphuric acid added to 1800 mL water.

- You might want to cut the agar blocks before the lesson. This saves time and makes the measurements more accurate. Phenolphthalein indicator makes the blocks pink in a basic solution (NaOH) and colourless when the acid diffuses into the agar.
- Surface-area:volume ratio is a difficult concept for many students. Having a tennis ball and large ball (basketball or netball) on hand can be useful. If you give students a ball of play dough they can mould it into different shapes and see that the same volume of dough can have very different surface-areas.

Teacher content information:

The surface area:volume ratio (SA: V) is an important concept in biology. As a cell or organism increases in size, its SA: V ratio decreases. You could ask students to calculate the surface areas and volumes of two cubes, one with sides 1 cm and the other with sides 3 cm.

Cube length (cm)	Surface area (cm ²)	Volume (cm ³)	SA:A ratio
1	6	1	6:1 = 6/1 = 6
3	54	27	54:27 = 54/27 = 2

The SA: V ratio has influence at the cellular, organ/system and whole organism levels:-

- Cells depend on the diffusion of materials (e.g. oxygen, carbon dioxide, nutrients, urea, and hormones) in and out of the cell via the cell membrane. This has implications for the maximum size of individual cells. Cells remain microscopic in size (high SA:V ratio) to allow fast diffusion of substances across the cell membrane, such that efficient metabolism occurs in the cell cytoplasm. In particular, red blood cells are flattened in shape (higher SA:V ratio) to allow more effective diffusion of oxygen in and out of the cells.
- At the organ/system level there are many good examples to illustrate this concept in the human body. Villi in the small intestine increase the surface area dramatically so that nutrients from digestion can pass into the blood efficiently, as do alveoli in the lungs, with regard to oxygen and carbon dioxide transport.
- At the whole organism level there are many examples of related species which are smaller in warm climates than in cold areas. The smaller animals are able to keep cooler in a hot climate by passing heat out through the surface of their smaller bodies, with relatively large SA:V, while the larger animals with smaller SA:V can conserve heat in colder climates.



Question - Why can some cells, such as eggs, be much larger than other cells?

Answer - The egg yolk of an unfertilised egg is a single cell. It comprises a little white dot (germinal disc) containing the nucleus and cytoplasm, sitting on top of the yellow yolk (fat and protein), which is really part of the cytoplasm. The egg cell or yolk is released from the hen's ovary and as it passes through the reproductive tract the egg white (albumen protein) is added plus a protective shell.

A fertilised egg is formed when a yolk meets a sperm cell in the reproductive tract. By the time the egg is laid the fertilised egg has undergone a few cell divisions to form an embryonic disc sitting on top of the yolk.

This dividing cell overcomes diffusion limitations by localising cell divisions to the small area of the embryonic disc. Most of the egg; the yolk and egg white, remains as a nutrition store. A short time later a network of blood vessels develops out from the germinal disc to better transport yolk and oxygen to the developing cell mass.

Lesson plan

- Step 1:** Introduce the lesson by revisiting **Activity 3.4** and discussing why the cell membrane is so important in keeping the cell alive and also brainstorm ideas about why cells are usually small.
- Step 2:** Explain the activity and then divide students into pairs or groups (3-4) to set up the experiment.
- Step 3:** While they are waiting for the results (about 15 minutes), students can design a table to record them.
- Step 4:** Before handling the blocks, ensure students rinse them well to remove acid and drain on a paper towel. They then cut the blocks to measure and record the diameter of pink left inside each block.
- Step 5:** Discuss the results, relating the amount of pink left inside each block to the efficiency of diffusion of materials in and out of the cell.
- Step 6:** Students work through the discussion questions, including the last on everyday situations. Play dough is useful here as a hands-on aid. Finally regroup the class to compare and discuss answers.

Suggested questions:

- How can you decide if this experiment is a fair test?
- What does the change in colour of the block tell us?
- What will happen to a cell if it becomes too large?
- Why can some cells, for example eggs, be much larger?
- If two cells have the same volume, which has a larger SA: V, a long skinny cell or a round cell?



3.6 Practical skills in biology

Lesson outcomes

At the end of this activity students will be able to:

- correct use of a microscope
- biological drawing
- drawing objects to scale.

Equipment list

Question 1 - MICROSCOPE STATION 1:

- label for station
- monocular microscope
- prepared slide e.g. plant stem TS.

Question 2: MICROSCOPE STATION 2:

- label for station
- small labels (for students to put their names on slides)
- monocular microscope
- microscope slide
- coverslip
- dropper bottle of methylene blue
- small beaker of water
- dropper/pipette
- probe
- scalpel
- piece of waterweed (e.g. Elodea) or onion slice.

Question 5b:

- specimen to draw, e.g. small leaf with serrated edges or a mollusc shell.

Each STUDENT will require:

- HB pencil
- eraser
- ruler
- **Sample test – Activity 3.6, Practical Skills in Biology (Pages 1-4).**



Things to consider:

- A marking scheme is available to teachers by request from *Science by Doing*. You may wish to print off a named copy for each student, so that you can mark microscope station 1. Attach the sheet to each student's test to record marks.
- **Microscope stations:** It is important to have enough stations to keep all the class working throughout the task. Rotate students to the stations while others work on questions.
- For a class of 24 or more students, run four stations for question 1 (needs marking on the spot) and six stations for question 2 (the slides can be labelled and marked later).
- **Specimens:**
 - For question 2, choose a pond weed with a single layer of cells e.g. Elodea or, if unavailable, onion skin can be used.
 - For question 5b, a small leaf with distinctive markings, or a mollusc shell, make good specimens, with details to test drawing accuracy. The specimen should be about 2 cm long so that it can be drawn to scale in the space available.
- Elodea is a noxious weed in some places but an alternative pond weed can be used.

Follow up: Any common problem identified by the task with microscope skills or drawing, could be reinforced in **Activity 4.1**.



4.1 Life in water

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Understanding cell structure and function has been possible due to the development of microscopes.
- Living organisms can be unicellular or multicellular.

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

Lesson outcomes

At the end of this activity students will be able to:

- use a microscope to observe microorganisms
- classify organisms based on the structure of their cells.

What ideas might your students already have?

Students should be able to recognise the differences between plant and animal cells (see **Activity 2.2**).

Key vocabulary:

Unicellular, multicellular, methyl cellulose.

Equipment list

Each PAIR will require:

- compound light microscope
- pond water sample
- microscope slides with wells
- coverslips
- dropper/pipette
- paper towel
- methyl cellulose dropper (optional)

Each STUDENT will require:

- **Notebook**

Things to consider:

- Pond water samples need to be collected with some sediment from the bottom, as this is where many multicellular organisms will be found. Stir the sample just before students begin the activity and instruct them to collect some sediment in their sample. Encourage students to share their findings and to get a new sample if they do not find many organisms in the first one.



- A drop of methyl cellulose added to the well on the slide will slow organisms down so that they can be viewed more easily.
- Cultures of pond organisms can readily be obtained from biological supply companies but local samples should give good results. (e.g. Southern Biological supplies mixed pond life culture (<http://www.southernbiological.com/mixed-pond-life-culture/>)).

Lesson plan

Step 1: Introduce the lesson by telling students they will explore the variety of life in pond water that we cannot see with the naked eye (microscopic organisms). Indicate that some of the organisms they might see are shown on pages 45-47 of the *Student Guide*, and that they can use these pages for identification, although it is not essential for them to know exactly what they are.

Step 2: Students use a slide with a well to collect a drop of pond water. Circulate to help locate organisms and ensure that students are following microscope procedures correctly.

Step 3: Students should draw/describe what they see in their **Notebooks**. If they follow the steps in the *Student Guide*, they should be looking at the type of cells and estimate the size of the organisms (refer back to **Activity 2.5**). They should try to identify the organism from the illustrations and think about their role in the pond community from the information given.

Step 4: Throughout the lesson have short show-and-tell sessions, as information is better shared when the organisms can still be viewed.

Suggested question/s:

- How can you tell if it is a plant or animal cell?
- What characteristics of life does the organism show?
- What characteristic of life does movement indicate?

Follow up:

You may like to develop this lesson further by asking groups or the whole class to draw up a food web for the pond. Brainstorm other organisms that would be present e.g. larger plants, fish, turtles, yabbies, birds etc.



4.2 Cells specialise

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.
- Growth and reproduction in living organisms involves cell division and specialisation.

Lesson outcomes

At the end of this activity students will be able to:

- describe a tissue as a group of similar cells carrying out a specific function
- recognise that the cells in tissues are specialised in structure to carry out their particular functions.

Key vocabulary:

Tissue, genes, specialised.

Equipment list

Each PAIR will require:

- monocular microscope
- microscope slides
- coverslips
- tissues/paper towel
- dropper/pipette
- iodine stain
- probe/toothpick
- scalpel
- selection of plant tissues e.g. rhubarb epidermis (peeled from the stalk), leaf epidermis, banana, potato
- prepared slides: TS woody stem (xylem) and of animal tissue e.g. muscle, bone, nerve tissue.

Each STUDENT will require:

- **Notebook**
- Internet access

Things to consider:

- **Making wet mounts** – the suggestions in the *Student Guide* all work well as wet mounts. Leaf epidermis needs to be stripped at an angle from the leaf. Many leaves work well (iris and lily leaves are recommended) but check that the epidermis separates at the torn edge before the lesson. A small piece of the edge can then be cut off and mounted on a slide. For banana and potato, very thin, small slices can be cut.



- Have prepared plant slides available (e.g. xylem tissue from a woody stem) if students have difficulty with the wet mounts.
- Investigate the virtual slides before the lesson, so that you can assist students to manipulate them.

Lesson plan

Step 1: Link this lesson to the previous discussion about cell division, colonies and multicellular organisms (**Activity 3.4**) to introduce the concept of a tissue and that multicellular organisms have many kinds of tissues for different jobs. The idea that all cells in an organism contain the same genes, but only some of these genes work in each type of cell, should also be introduced.

Step 2: Students work in pairs to prepare wet mounts of plant tissues. Reminders about microscope procedures, staining technique and class drawing rules are important here. Students should follow the steps indicated and draw the cells. Quicker students may repeat this with a second wet mount. If you are using leaf epidermis, point out the stomatal openings, as these will be referred to in **Activity 4.3**.

Step 3: Students investigate prepared animal tissue slides in the same way.

Step 4: Students investigate digital slides of tissues. Ensure **Notebook** questions are read before and during the observation of the slides, and that students know how to zoom in on cells to view the details more clearly. You might want to discuss some questions as the activity progresses to check they are on track.

Suggested question/s:

- What do all tissues have in common?
- How do you think tissues form?
- Give an example of how a cell type is specialised for a particular role.
- Why do multicellular organisms need specialised cells?



4.3 Putting it all together

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.
- Organ systems of organisms in different environments show adaptations for survival in that niche.

Lesson outcomes

At the end of this activity students will be able to:

- describe the relationships between cells, tissues, organs and body systems
- discuss why multicellular organisms require cell specialisation to survive.

What ideas might your students already have?

Students should have some idea of the organisation of the human body and be familiar with the tissues as the first step in the organisation of body cells (**Activity 4.2**).

Key vocabulary:

Organ, body system, stomach, muscle, vascular tissue, digestive system, stomatal opening, photosynthesis, xylem, phloem, bark, cambium, transpiration, diffusion, carbon dioxide, oxygen, chloroplast, chlorophyll.

Equipment list

The CLASS will require:

- internet access

Each STUDENT will require:

- **Notebook.**

Teacher content information:

- **Leaf adaptations** - the main adaptations students should see are:
 - stomatal openings on the leaf surface (epidermis) to allow gas exchange
 - spaces between the cells inside the leaf to allow gases to reach all cells
 - palisade (food-making) cells are on the outside of the leaf to enable them to obtain maximum light
 - veins (vascular tissue) are bundles of long tubes to carry water and nutrients around the plant.
- **Plant stem** – most of the trunk of a tree is composed of dead tissue. The only living cells are the cambial cells just under the bark where new xylem and phloem tissues are produced.
- **Photosynthesis** – The *Find out more* section introduces students to the structure of the leaf and demonstrates the movement of carbon dioxide into the leaf and oxygen out of the leaf. Sunlight energy for photosynthesis is collected by the pigment chlorophyll, located in the chloroplasts of palisade cells.



- **Transpiration** - the process of water movement through a plant and its evaporation from the leaves.

As you teach:

Step 1: Begin with page 50 of the *Student Guide* on the screen to discuss the need for organisation in multicellular organisms and to introduce the terms organ and system, using the examples provided.

Step 2: Students can form pairs to view the click-on representations of the human digestive system and a leaf of a flowering plant.

Step 3: Direct them to the *Find out more* to learn more about plants. They should follow the **Notebook** guide to develop an understanding of the similarities in organisation and specialisation of the tissues and organs in each case. Students may need guidance here, particularly in interpreting the role of plant tissues.

Step 4: Reform the class for an overview at the end of the lesson.

Suggested question/s:

- What is the role of the ... system?
- Can you name an organ in the ... system?
- What tissues would be needed in the ...?
- Why do multicellular organisms need any kinds of specialised cells to survive?

Follow up:

The understanding of the concepts in this activity can be assessed in **Activity 4.8**.



△ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

4.4 Organs and systems

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- identifying the organs and overall function of a system of a multicellular organism in supporting the life processes
- describing the structure of each organ in a system and relating its function to the overall function of the system.

Lesson outcomes

At the end of this activity students will be able to:

- discuss the role of body organs and systems in maintaining effectively functioning organisms
- recall the main parts and functions of the digestive system.

What ideas might your students already have?

Students should understand the need for organs and body systems in multicellular organisms from previous lessons in this unit.

Key vocabulary:

Saliva, carbohydrates, proteins, oesophagus, peristalsis, stomach, chyme, pancreas, liver, small intestine, nutrients, large intestine, constipation, enzyme, salivary gland, nutrient, faeces, anus.

Equipment list

- **Each STUDENT will require:**
- Internet access
- **Notebook.**

Things to consider:

The interactive In Digestion follows the passage of food through the human body. Students select three foods and drinks and decide how to digest them (e.g. choose to chew, to add saliva to the mouth or add gastric juices to the stomach), and watch how the body reacts to changes. It provides feedback and shows consequences, e.g. vomiting. Problems such as choking, stomach ache, gas and diarrhea are explained.

- A five-page glossary can be printed from the interactive In Digestion (click on the 'C' in the bottom left corner).
- The optional *Find out more* section on skin is a valuable introduction to **Activity 4.5 New skin**, which looks at the work of burns surgeon and 2005 Australian of the Year, Professor Fiona Wood. It has sections on:
 - o structure and functions of the skin
 - o burns and their treatment
 - o skin cancer (14 minute Australian video).



If time is short, quicker students could look at different sections and report back to the class. You might like to set some parts for homework.

As you teach:

- Step 1:** Project the **Student Guide** onto the class screen and ask students to think-pair-share answers to the question 'Could we live without one of these body systems?'. Invite class responses and develop students' understanding that all systems are crucial to survival and that they work together to support each other.
- Step 2:** Direct students to explore the interactive on the Digestive System. Allow a good 20 minutes for this activity as students learn more as they delve into this non-linear program. Invite more able and motivated students to choose the 'Find out more' options when they appear. Encourage students with poor literacy to use earphones and to click on new words to hear their pronunciations.
- Step 3:** Invite students to complete all the **Notebook** questions, before discussing as a class.
- Step 4:** Ask students 'Which is the largest organ in the body?' and invite them to explore the *Find out more* section (see *Things to consider*).



4.5 New skin

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.
- Growth and reproduction in living organisms involves cell division and specialisation.

ACSHE134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.

ACSHE226 Science knowledge can develop through collaboration across the disciplines of science and the contributions of people from a range of cultures.

ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

Lesson outcomes

At the end of this activity students will be able to:

- discuss the importance of science in advancements in medicine
- develop literacy skills by reading, viewing and listening to media forms to analyse and communicate information
- develop an understanding that perseverance is necessary to make significant achievements.

What ideas might your students already have?

Students might have heard about the Bali bombings but they would not remember the events at the time.

Equipment list

Each **STUDENT** will require:

- Internet access
- **Notebook.**

Things to consider:

This lesson focuses on literacy and second-hand data research. A video transcript of the Fiona Wood video may assist hearing impaired students.

Lesson plan

Step 1: Explain this is a research lesson, briefly introduce the topic and then give students quiet time to read the information in the *Student Guide*, including the article on journalist, Cynthia Banham (pages 53 - 54). Then discuss with the class what they have learnt so far about Prof Wood.

Step 2: As a class, watch the interview with Prof Wood (10 min) and read the article *Holiday haven turns to hell on Earth*.

Step 3: Form groups of 3-4 to discuss the **Notebook** questions. You could then give each group one question to report on to the class.



Step 4: Finish with a class discussion, after groups have reported. This will enable differences in ideas to be expressed or the discussion to be extended.

Suggested questions:

- What are some functions of the skin?
- What effect would severe burns have had on a person's life before recent medical advances?



⚠ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

4.6 Let's look inside

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Organ systems of organisms in different environments show adaptations for survival in that niche.

Lesson outcomes

At the end of this activity students will be able to:

- describe the main structures of a squid and their functions.

Key vocabulary:

Tentacle, ink sac, gills, gladius, mantle, beak.

Equipment list

Each **GROUP** will require:

- squid
- dissecting or baking tray
- sharp scissors
- bowl or sink filled with water
- latex gloves (optional)
- lemon (optional)
- plastic bag.

Things to consider:

- There is a short video (1 min 10 sec) of the giant squid which could be used as an introduction to this activity at <http://science.discovery.com/video-topics/strange-creatures/fyi-giant-squid.htm>
- Students love dissections, but some can be squeamish. Providing gloves and hand-washing facilities in the laboratory make the experience more pleasant for all. Provide a garbage bag, so that the remains can be correctly disposed of. Students should rinse all equipment and ensure sinks are clean to avoid odour later.
- Squid is easily obtained from a fishmonger. However, you may prefer one of the alternative activities below:
 - Fish dissection CSIRO <http://www.csiro.au/helix/sciencemail/activities/fishliver.html>
 - Pluck demonstration (heart and lungs) <http://www.getbookee.org/pluck-dissection-lab/>
 - Rat dissection demonstration.
 - There are many videos available showing the dissection of various body systems. e.g. <http://www.youtube.com/watch?v=A7SrFNIRots>
- Be aware of latex allergies and fainting risk when dissecting.



Teacher content information:

Squid are cephalopods in the phylum Mollusca. They show relatively high intelligence, like octopuses. They are able to communicate and some hunt in groups.

Like all other cephalopods, squid have a distinct head, bilateral symmetry, a mantle and tentacles. Squid have eight tentacles arranged in pairs and two, usually longer, tentacles. They are strong swimmers and certain species can 'fly' for short distances out of the water. The main body mass is enclosed in the mantle, which has a swimming fin along each side; however, these are not the main method of locomotion in most species.

The skin is covered in chromatophores, which enable the squid to change color to match its surroundings, making it practically invisible. The underside is also almost always lighter than the topside, to provide camouflage.

Under the body are openings to the mantle cavity, which contains the gills, and openings to the excretory and reproductive systems. At the front of the mantle cavity is the siphon, which the squid uses for movement by jet propulsion, sucking water into the mantle cavity and expelling it from the siphon in a jet. The direction of the siphon can be changed, to alter the direction of travel.

Inside the mantle cavity is the visceral mass, covered by a thin membrane. Below are all the major internal organs. The ink sacs are located behind the gills. A cloud of dark ink is released when the squid is startled. This is primarily a defence against predators.

Lesson plan

Step 1: Introduce the lesson with a discussion about what the students know about squids, using page 55 of the *Student Guide* as stimulus.

Step 2: Go through the steps on page 56 before the students collect their squid. Stress the importance of looking at the whole organism and finding all the features before any incision.

Step 3: Students do the dissection in groups of 3-4. Circulate to assist in finding parts.

Step 4: Leave plenty of time to clear up. Make sure all equipment is rinsed and sinks are clean.

Suggested questions:

- What do we eat that is made from squid?
- How do you think a squid moves?
- What is the purpose of the ink the squid produces?
- Do squid have backbones?
- How do they support their bodies?
- How many roles can you think of for tentacles?

Follow up:

For enthusiasts, there is a video of the public dissection of the giant squid: this is over an hour long and includes some live footage of cephalopods- <http://museumvictoria.com.au/about/mv-news/2008/giant-squid-public-dissection-at-melbourne-museum/>



4.7 Plant plumbing

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Organ systems of organisms in different environments show adaptations for survival in that niche.

Lesson outcomes

At the end of this activity students will be able to:

- describe how water and nutrients are transported to all parts of a plant
- discuss the role of stomata in plant metabolism.

Key vocabulary:

Vascular tissue, xylem, phloem, transverse section, longitudinal section, transpiration, stoma (or stomate, plural is stomata), guard cell.

Equipment list

Each GROUP will require:

- 1 leafy celery stem left in coloured water overnight
- scalpel
- monocular microscope
- slides and coverslips
- dropper/pipette
- fresh leaf
- clear nail polish.

Things to consider:

- **Cutting sections of celery** – the clearest sections are the thinnest. The best tip is to cut a small sliver, as thin as possible, rather than trying to cut a complete section.
- **Stomata slides** – a thin layer of nail polish 1 - 2 cm² works best. It should be applied in a single process, as if to nails. The strip will peel off easily when dry and leaves a clear imprint of the epidermal cells and stomatal openings. Leaves with slightly waxy surfaces are best. Any citrus leaf works well. Others to try are camellia, gardenia, hydrangea, eucalyptus and ivy. Some good compare and contrast activities with stomata are described on the downloadable Plant Energy Biology activity sheet 'How to perform a stomata peel' (http://plantenergy.edu.au/news/media/Stomata_Peel.pdf).



Teacher content information:

Celery vascular tissue can be seen clearly using red food dye; showing the water movement through the xylem vessels.



In longitudinal section, the vascular bundles can be seen as long tubes



Lesson plan

- Step 1:** Begin by discussing the vascular tissue as a transport system, explaining the TS and LS images on page 57 of the *Student Guide*.
- Step 2:** Students work in pairs to make wet mounts of celery stem, following the instructions. Stress that a thin, small section, not a big, fat slice is needed. Reiterate the drawing rules and ask students to draw a few cells (large). Discuss what they can see and how this helps the plant to survive.
- Step 3:** Paint a thin layer of clear nail polish onto both surfaces of a leaf (see *Things to consider*). Peel off the dry skin and mount on a slide to view the epidermis and stomata. Students should be able to make a labelled drawing of a stoma and surrounding epidermal cells, using the diagrams on page 58 as a guide.
- Step 4:** Conclude by discussing the questions in the *Student Guide* and comparing the levels of organisation of plants with that of animals.

Suggested questions:

- What shape are xylem vessels?
- What structures in our own bodies does vascular tissue remind you of?



4.8 How does a multicellular organism function?

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.

Lesson outcomes

At the end of this activity students will be able to:

- describe the relationship between cells, tissues, organs and systems.

What ideas might your students already have?

Students should have mastered the concepts involved. This activity can be used as a formative assessment task.

Equipment list

Each **GROUP** will require:

- **Activity sheet 4.8 Body system cards** (in *Teachers Guide*)
- scissors
- butcher paper,
- coloured pencils/markers.

Things to consider:

This activity can be used as an informal **formative assessment**. It works best if organised as a competition between the groups. You could give a small prize to the winning group, on the condition they are quiet until every group has finished. They can be planning their poster quietly while they wait. The system cards can be mounted on card and laminated for future use.

Lesson plan

Step 1: Divide the class into groups of 3-4, preferably spread out and around tables. Explain the task, asking them to work quietly so that other groups cannot overhear their discussion.

Step 2: Students follow the task instructions. When finished, they put their hands up and wait quietly for their work to be checked. Review any problems.

Step 3: Groups can begin their poster when ready, choosing one body system to illustrate. It is useful for display purposes to ensure each body system has been chosen by at least one group.

Step 4: Finish with a **Gallery walk** for the students to share their work.



5.1 Is death a part of life?

Curriculum links

ACSSU149 Cells are the basic units of living things; they have specialised structures and functions.

KEY IDEAS

- Cells are the basic units of living things.

Lesson outcomes

At the end of this activity students will be able to:

- develop literacy by interpreting visual media
- describe the death and renewal of cells as part of the maintenance of healthy organisms.

What ideas might your students already have?

Students have some background on the function of the intestines in digestion from **Activity 4.4**, and of cell division and differentiation from Parts 3 and 4 of this unit.

Key vocabulary:

Stem cell, lysosomes, necrosis, programmed cell death, toxin, tumour, villi, goblet cell, secretory cell, absorption cell, transitional cell, mucous, enzyme.

Equipment list

The **CLASS** will require:

- Internet access.

Each **STUDENT** will require:

- Notebook.**

Things to consider:

- Familiarise yourself with the animation before the lesson. See the brief synopsis below.
- Success depends on a good introduction so students understand terms and images before they watch the animation.
- A good strategy is to watch the animation once, then review in short sections, pausing to interpret each part. If students make notes as they go, they will be well placed to write their story afterwards.

Teacher content information:

Life and Death of an intestinal cell

- The animation begins by zooming in on the villi of the small intestine to focus on the cross-section of a single villus.
- Some cells are seen to die, while stem cells at the base of the villus divide by mitosis to produce new cells, which become transitional cells. These then divide to form more transitional cells.



The next sections (3-5) deal with differentiation of the transitional cells into specialised cells

3. Transitional cells differentiate into secretory cells, which produce digestive enzymes (shown as small coloured spheres).
4. Transitional cells differentiate into absorption cells. These are seen to take in larger molecules of nutrients and then pass them as small molecules into the lacteals and capillaries serving the small intestine, to be passed around the body and delivered to all cells.
5. Transitional cells differentiate into mucous cells, which spread a layer of mucous inside the intestine.
6. Programmed cell death is shown, with some cells disappearing (being reabsorbed) and the empty spaces being filled by adjacent cells.
7. The formation of a tumour is shown by repeated divisions of cells that have not differentiated correctly.
8. Necrosis is illustrated by the invasion of bacteria. These produce toxins, which cause the cells to change colour and eventually explode. The spaces these cells occupied remain, to illustrate tissue damage.

Lesson plan

Step 1: Begin the lesson with a discussion of cell death and renewal through mitosis and cell specialisation, using the examples in **Activity 3.4** and on pages 62-63 of the *Student Guide*.

Step 2: Before students view the animation, go through the types of cells they will see in the villi; this will help during the viewing.

Step 3: The animation can be watched as a class or in groups. Afterwards, students should review the animation in short sections and make notes on what is occurring. You may need to give clues.

Step 4: Students write a story about what is happening. Circulate during this time. You may ask a few students to read out their stories.

Step 5: Reform the class. Ask what they have learnt and how the animation helped their understanding. Discuss the ways animations are not 100 per cent accurate.

Suggested questions:

- Describe what is happening here?
- What type of cell is this?
- What do you think its role is?
- How is the cell changing?



5.2 A scientific breakthrough

Curriculum links

- ACSHE134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- ACSIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.

Lesson outcomes

At the end of this activity students will be able to:

- discuss the importance of scientific methodology in advancements in medicine
- develop literacy skills by reading, viewing and listening to media forms to analyse and communicate information
- develop an understanding that perseverance is necessary to make significant achievements.

Key vocabulary:

Gastric, ulcer, gastritis, Helicobacter pylori, endoscope, antibiotic, Nobel Prize for Medicine.

Equipment list

Each **GROUP** will require:

- butcher paper and marking pen.

Each **STUDENT** will require:

- internet access
- **Notebook.**

Things to consider:

- It is important to familiarise students with the vocabulary needed to follow the interview with Prof Marshall. This can be done by studying the introductory material in the *Student Guide* before moving on to the video.

Lesson plan

Step 1: Begin with a discussion of the background information on page 64 of the *Student Guide* to ensure students are familiar with the terms relating to Prof Marshall's work.

Step 2: Students can watch the interview as a class (10 min). You could pause at intervals to ensure students have understood the concepts.

Step 3: Students should read the newspaper article before attempting the **Notebook** questions. A class discussion about answers to the **Notebook** questions works well here.

Step 4: For the final activity, students form groups of 3-4 to compare on butcher paper the life and work of Prof Marshall with that of Prof Wood. Students can view other groups' work as a **Gallery walk**. Interesting points can then be shared with the class.



Suggested questions:

- Why was the endoscope an important breakthrough in medical technology?
- Why is it important that people who do medical research are not connected to drug companies?
- What are the essential things that must be considered when designing a controlled experiment?



△ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

5.3 Why the difference?

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Organ systems of organisms in different environments show adaptations for survival in that niche.

Lesson outcomes

At the end of this activity students will be able to:

- describe the relationship between mouthparts and diet
- explain how the skeletons of animals make them well adapted to their lifestyles

Key vocabulary:

Herbivore, carnivore, omnivore, canine, molar, incisor, proboscis.

Equipment list

Each GROUP will require:

- butcher paper and marking pen.

Each STUDENT will require:

- Notebook.**

Lesson plan

Step 1: Introduce the lesson by brainstorming with the class what they remember about feeding relationships in communities and names given to organisms with particular diets. Students can then look at the examples of animals on pages 65 – 66 with different diets and discuss the adaptations of their mouthparts, tabulating their ideas in their **Notebooks**.

Step 2: On page 66, consider examples of animals with different lifestyles. Students should follow the directions to look at the adaptations and similarities and differences between the skeletons shown. Ask each student group to talk to their poster and invite other groups to add any ideas they gain from other groups to their posters.

Suggested questions:

- What do you notice about the teeth of an animal that eats...?
- What kind of teeth are missing in herbivores?
- What kinds of teeth are most prominent in carnivores?
- What are molars used for?
- What is similar about all these skeletons?
- What is different about the skeleton of a...?
- Can you explain how this helps the animal to survive?



5.4 Reproductive strategies in animals

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Sexual reproduction involves fusion of gametes and cell division, whereas asexual reproduction involves mitosis in a single organism.

Lesson outcomes

At the end of this activity students will be able to:

- recognise that animal reproductive strategies are related to their environment and the complexity of the organism
- summarise data from secondary sources and use scientific understanding to identify relationships and draw conclusions.

What ideas might your students already have?

Students should have an understanding of mitosis and its role in asexual reproduction from **Activity 3.4**.

Key vocabulary:

Fertilisation (external and internal), sexual reproduction, asexual reproduction, binary fission, mitosis, budding, gametes, mating, hermaphrodite, life cycle, larva, chrysalis, pupa.

Equipment list

Each GROUP will require:

- Internet access
- butcher paper and marking pen.

Each STUDENT will require:

- Notebook.**

Things to consider:

- A guided class discussion is a good start, using the material in the *Student Guide*. Ensure students grasp the idea of the trend from asexual to sexual reproduction in invertebrates and strategies leading to increased survival rates of young in vertebrates. Revision of the vertebrate groups is recommended. The link between complexity of the organism and sophistication of reproduction should be noted.
- Questions on the video clips could be discussed in groups. The study of an Australian animal not covered in **Activity 5.4** could be a group activity, with a lucky-dip to ensure of a good range of organisms. Use examples such as koala, wombat, dingo, ringtail possum, Tasmanian devil, platypus, kookaburra, brush turkey, bower bird, rosella, brown snake, blue tongue lizard, goanna, crocodile, barramundi, long-finned eel.
- Butcher paper and marker pens are useful for groups to answer **Notebook** questions 6 and 7.



Teacher content information:

- Notes on the video clips:
 - Christmas Island crab migration** (2' 48") - annual migration from rainforest to ocean to lay eggs.
 - Reproduction in locusts** (2' 30") - swarming of plague locusts explained in an Australian context, including egg laying.
 - Echidna hatching** (1' 2") - development of a baby echidna from an egg.
 - Reproduction in a marsupial mammal, the kangaroo** (3' 17") - birth of a baby kangaroo, migration of the joey to the pouch, suckling and development of independence.
 - Reproduction in placental mammals** (3' 27") - birth of a wildebeest demonstrating the advantages of mature birth, with the young animal standing up and running soon after.
- Here is a completed table of reproductive characteristics of the animals. An example of the frog and platypus life cycles is also included. This information is also available as **Activity sheet 5.4** Suggested answers (in *Teacher Guide*) if you would like to print/project it for the class.

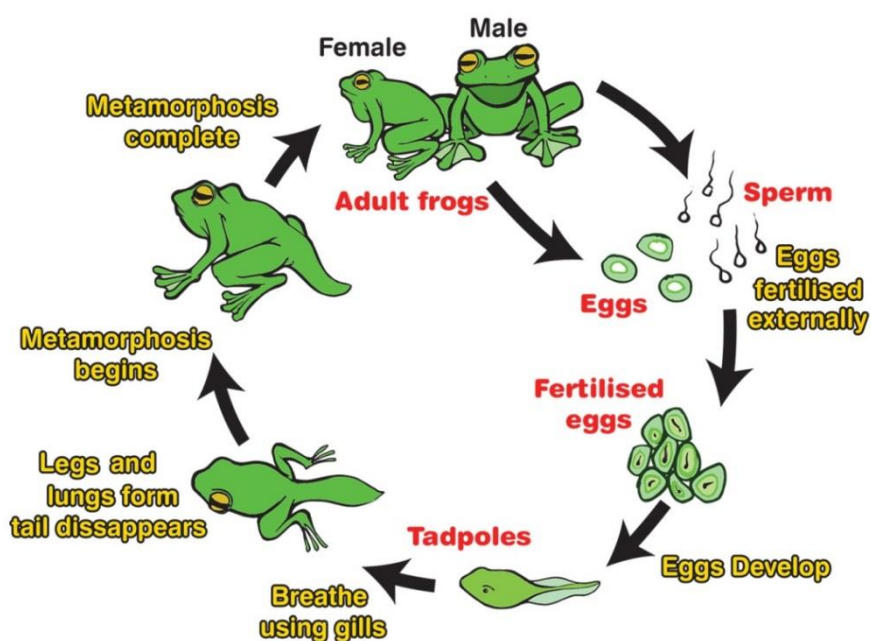
Summary of reproduction in animals

Animal	Type of reproduction	Where eggs are laid/ fertilised	Number of eggs/ live young	Survival rate	Care by parents
Paramecium	Asexual*	N/A	N/A	N/A	N/A
Hydra	Asexual*	N/A	N/A	N/A	N/A
Coral	sexual	Randomly in water	Thousands of sperm and eggs released annually	Less than 1%	N/A
Slug	sexual	Externally as they leave the female's body	About 200 eggs laid (varies with species)	About 20% hatch but only a few of these reach maturity	N/A
Yabby	sexual	Externally in water as they leave the female's body	100-300 eggs laid	About 10% survive to maturity	Female carries under tail until eggs hatch, then released into the water (some attach to parent in early stages)

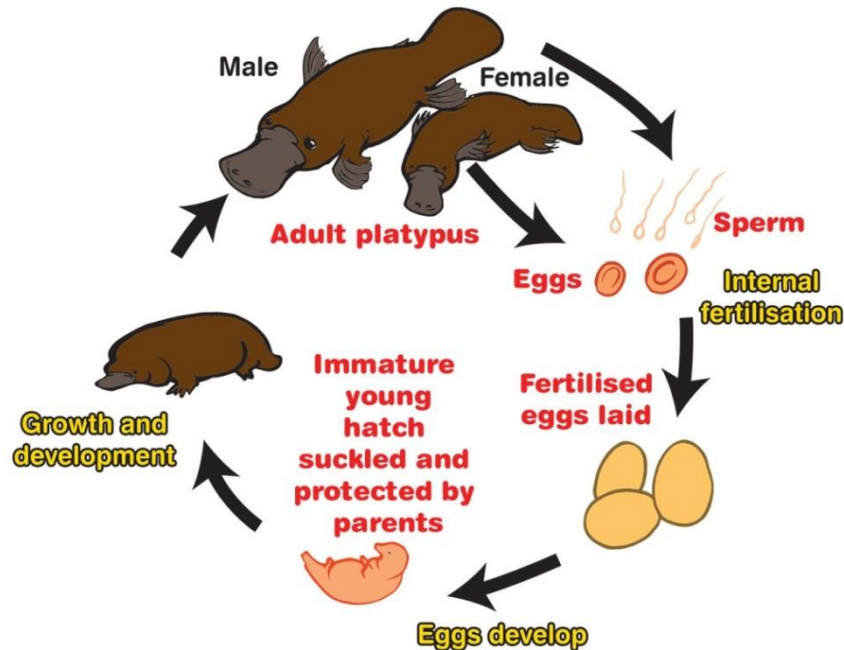


Animal	Type of reproduction	Where eggs are laid/ fertilised	Number of eggs/ live young	Survival rate	Care by parents
Fish	sexual	Externally in water as they leave the female's body	Thousands of eggs laid	Only a few survive to maturity	Usually no care of eggs or young by parents
Frog	sexual	Externally in water as they leave the female's body	250 -1000	About 1% survival to maturity	Usually no care of eggs or young by parents
Turtle	sexual	Internally, eggs laid on land	100	About 1% survive to maturity	No care of eggs or young by parents
Bird	sexual	Internally	2-6 eggs laid	1 or 2 per clutch survive to maturity	Eggs and chicks cared for by parents
Dog	sexual	Internally	3-10 puppies born	In the wild only 2 likely to survive	Young suckled by mother and stay with parents until mature

FROG LIFE CYCLE



PLATYPUS LIFE CYCLE



As you teach:

- Step 1:** Begin with a guided discussion of the reproductive features of the animals, encouraging the students to look for trends in both invertebrate and vertebrate reproduction. Form groups of 3-4 to draw up a table to help answer questions in the *Student Guide*.
- Step 2:** Students watch and discuss the video clips in groups or as a class, first checking the **Notebook** activities.
- Step 3:** In groups of 3-4 discuss the **Notebook** questions and investigate an Australian animal not mentioned in this activity. A short report to the class can follow.
- Step 4:** Students study the interactive butterfly life cycle and adapt this format to draw life cycles for a frog and platypus. This can be planned in groups and drawn on butcher paper for class display and discussion, to compare these three cycles.

Suggested questions:

- Why do you think paramecium and hydra usually reproduce asexually, while the other invertebrates shown use sexual reproduction?
- What strategies do animals that lay a large number of eggs have in common?
- Why is internal fertilisation an advantage on land?
- Which animals care for their eggs or young?
- What do these animals have in common?



5.5 Plants reproduce too!

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Sexual reproduction involves fusion of gametes and cell division, whereas asexual reproduction involves mitosis in a single organism.

Lesson outcomes

At the end of this activity students will be able to:

- describe the role of a flower in the sexual reproduction of plants.

What ideas might your students already have?

- Students often do not think of plants as organisms that reproduce sexually.

Key vocabulary:

Compound flower, petal, sepal, stigma, style, ovary, ovule, anther, filament, stamen, carpel, pollen, pollination, fertilisation, pollen tube.

Equipment list

Each STUDENT will require:

- internet access
- Notebook.**

Each PAIR will require:

- flower specimens
- scalpel
- cutting board
- hand lens
- microscope
- microscope slides
- cover slips
- dropper/pipette.

Things to consider:

- The practical session is best done at the end of this activity, when students are familiar with the parts of a flower and the many adaptations that occur. This enables better interpretation of different flowers.

Teacher content information:

- Notes on the video clips:

Woodland timelapse (1' 05") - dramatic introductory video compressing six-months' plant growth and development into one minute. It emphasizes plants as living things, exhibiting the characteristics of life.



Pollination and fertilization (0' 54") - the distinction between pollination and fertilization is clearly demonstrated with a large model flower. Microscopic footage of a germinating pollen tube and fusion of the male and female nuclei.

Wasp Flowers (2' 57") - orchids mimic a female wasp to attract the male wasp for pollination.

Pollination Prisoners (4' 00") – see the dead horse arum lily mimicking a rotten corpse near gull colonies. This attracts blowflies for pollination.

Bees and Balsam (1' 36") - bees pollinating balsam flowers.

Hay fever and pollen (2' 50") - the link of pollen to hay fever, the structure of grass flowers and their wind assisted pollination.

Dissecting Flowers:

Suggest students examine simple flowers, e.g. clivia, lilies or single azaleas, then those more difficult to interpret. If using compound flowers, ask students to take a single flower from the head.

Other interesting flowers are violets, agapanthus, sweet peas, daffodils or jonquils and many others, depending on the season.

The flowers of native plants can be tricky, as many have no petals, with the stamens forming the coloured parts. However, it is worth having some on hand to discuss the adaptations, e.g. bottlebrush (callistemon), grevillea, wattle (acacia).

Pollen grains: These can be easily shaken from stamens or removed attached to sticky tape and stuck directly onto a slide. Students should see that the pollen of different flowers is distinctive, making it a feature used in detecting the presence of plant species, e.g. in forensic samples.

Lesson plan:

- Step 1:** Introduce this activity with a class viewing of Woodland timelapse, followed by a discussion of how it highlights the characteristics of plants.
- Step 2:** Discuss different types of flowers using the illustrations in the *Student Guide*. Review the details of flower structure and give students time to learn the parts and their function in pairs (5 - 10 minutes).
- Step 3:** Introduce pollination and fertilisation by reviewing the diagrams in the *Student Guide* before showing the class the video clip to consolidate these ideas.
- Step 4:** Students can watch the video clips on different adaptations for pollination, either in small groups or as a class. They should then individually do the **Notebook** questions and the click-and-drag exercise on flower structure.
- Step 5:** The practical session involves dissecting flowers and observing pollen under the microscope. Students should follow the instructions in the *Student Guide*. Direct them to the most straightforward flowers first. Stop the class at convenient points to discuss problems with interpretation and involve students in discussing adaptations.

Suggested questions:

- What drives the yearly cycle of plant growth?
- Can you suggest why there is such a wide variety of flower structure?
- Why do many flowers grow in clusters?
- Where on a plant would you be likely to find wind-pollinated flowers?



5.6 But there's another way!

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Sexual reproduction involves fusion of gametes and cell division, whereas asexual reproduction involves mitosis in a single organism.

Lesson outcomes

At the end of this activity students will be able to:

- describe asexual reproduction as cloning to produce individuals genetically identical to the parent plant
- describe some methods of asexual reproduction in plants
- discuss the differences and advantages / disadvantages of sexual and asexual reproduction.

What ideas might your students already have?

Students probably do not realise that many plant species can reproduce both sexually and asexually.

Key vocabulary:

Rhizome, bulb, runner, tuber, clone.

Equipment list

The CLASS will require:

- selection of labelled plant reproductive organs spread about the room.

Each STUDENT will need:

- Notebook.**

Things to consider:

Asexual reproduction practical

Set up stations around the room with labelled specimens. Suggestions are:

Runners - strawberry, violet, kikuyu grass, buffalo grass, Spinifex

Tubers - potato or dahlia

Bulbs - onion

Rhizomes - iris, Canna lily

Comparing sexual and asexual reproduction - a table with suggested answers is in *Teacher Content Information*.

Teacher content information:

Sexual v asexual reproduction: When growing conditions are good, e.g. warm with adequate water, plants that can reproduce asexually will often do so vigorously. However, when plants are stressed, e.g. by extreme heat or drought, this will often stimulate flower production. The seeds produced can survive adverse conditions and germinate when the conditions are more favourable.



Underground parts, e.g. bulbs, formed by asexual reproduction in good growing conditions, are also a means of surviving adversity. These plants will also produce flowers at the end of the growing season. The seeds are a method of dispersal, as well as providing genetic diversity, whereas bulbs will provide new 'clone' plants in the same location.

Table comparing sexual and asexual reproduction

Feature	Sexual reproduction	Asexual reproduction
Similarity to parents	All different as they develop from fertilised eggs, so have a mixture of inherited material from both parents	Identical genetically to parents (clones) as formed by mitosis
Time taken	Development is a relatively slow process	Many offspring can be formed quickly
Involvement of other individuals	Fertilisation of gametes required, which usually requires another individual	Is carried out by a single individual
Survival	Seeds can survive adverse conditions e.g. drought	Reproductive parts can survive adverse conditions e.g. cold winters
Amount of energy used	Takes a lot of energy to mate and produce young	Little extra energy required
Adaptation to the environment	New combinations of genes may be less favourable, but occasionally offspring are better adapted than their parents	If parent is well adapted to living conditions, then offspring will be too

Lesson plan:

Step 1: Begin by introducing the idea that plants can reproduce asexually and link to previous understanding of mitosis producing genetically identical cells.

Step 2: Form groups of 3-4. Rotate the groups around the stations with labelled reproductive parts (see notes) to complete the table and discussion questions. Bring the groups together to compare answers.

Step 3: Students work in groups to draw up a table of comparison between asexual and sexual reproduction, using what they have learnt from visiting the stations and the stimulus material on pages 72-73 of the *Student Guide*. You may need to help students to understand the meaning of the features. Finish by discussing the main similarities and differences.

Suggested questions:

- Under what sort of conditions are plants likely to reproduce asexually?
- When might plants reproduce sexually?
- Which type of reproduction would require more energy? Why?
- Which type of reproduction would produce most offspring in a stable environment?
- Which method might be an advantage if the conditions are changing



5.7 Presenting and interpreting experimental results

Curriculum links

- ACSYS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- ACSYS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- present data in a table
- construct and interpret line graphs
- draw valid inferences and conclusions from experimental data.

What ideas might your students already have?

Students will have some experience with these tools of the scientific method, but may not be able to work independently in this area.

Key vocabulary:

Independent variable, dependent variable, axes, line of best fit.

Equipment list

Each STUDENT will need:

- graph paper
- ruler
- sharp HB pencil
- rubber
- **Notebook.**

Things to consider:

- This activity can be done at any stage in the unit, as the need for organising data, graphing and drawing conclusions are required, for example, before **Activities 1.3** and **3.3**. It is important to revise at this stage to prepare for writing a report on bean growth (**Activity 5.8**).
- Clarify the terms *independent* and *dependent variables* at each stage. Encourage students to put the independent variable in the left column of their table, as this makes transfer of the data to a graph more straightforward.
- Sample tables and graphs are to be found in **Activity sheet 5.7 Presenting and interpreting experimental results**.

Lesson plan:

Step 1: After discussing why tables and graphs are so important to scientists, students follow instructions to draw a table of *counting snails* data from a written paragraph.



Step 2: Review the graphing instructions on page 75 as a class before students construct graphs from the two tables of data on page 74 and 75.

Step 3: Discuss what information graphs can tell us, how to describe trends and what other things should be considered when writing a scientific report on an experiment. Students then answer the questions on page 75. Conclude by asking some students to give their answers and discussing responses.

Suggested questions:

- What is the independent variable
- What is the dependent variable
- Where should you put the units in a table/graph?
- On which axis do you plot the independent/dependent variable?
- How can you decide if your experiment is a fair test?
- What can you do to make an experiment reliable?
- Is this a valid inference/conclusion from this data?

Follow up: The skills in this activity can be tested in **Activity 5.8**



△ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

5.8 So how did it grow?

Curriculum links

ACSSU150 Multi-cellular organisms contain systems of organs carrying out specialised functions that enable them to survive and reproduce.

KEY IDEAS

- Multicellular organisms contain systems of organs that carry out specialised functions to enable them to survive and reproduce.
- Growth and reproduction in living organisms involves cell division and specialisation.

AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identify improvements.

AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

Write a scientific report on a first hand investigation including:

- presenting results in a table.
- constructing and interpreting a graph.
- drawing valid conclusions from observations.

Equipment list

Each STUDENT will need:

- instructions for task and marking scheme
- **Notebook** with information collected as the beans grew
- graph paper
- lined paper to write report (or this could be done as a computer task).

Things to consider:

- A suggested task and marking scheme for the written report is provided in **Activity sheet 5.8 Bean growth report rubric**.

Make sure your class has practised the skills of tabulation, graphing and scientific report writing before attempting this task.



5.9 Sample Test

A sample summative test for Units 1 - 5 (editable version) and marking scheme are available to teachers by email request from *Science by Doing* (sbd@science.org.au). In your email request please include your school (or university) and contact details.

Note - *Science by Doing* provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. *Science by Doing* encourages teachers to modify the items to suit individual teaching programs.



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